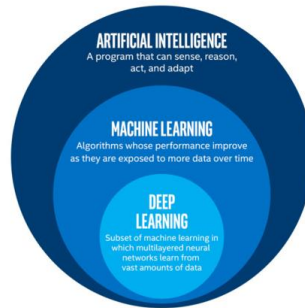


Inteligência Artificial

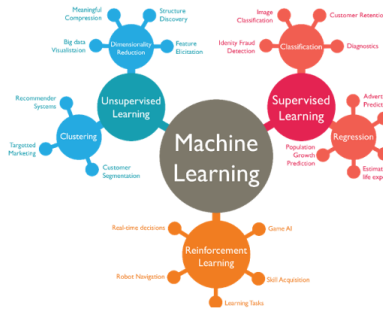
Machine Learning

Agenda

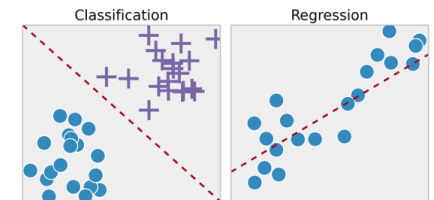
Artificial Intelligence



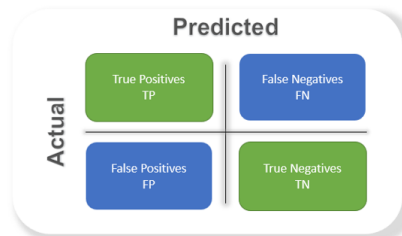
Machine Learning



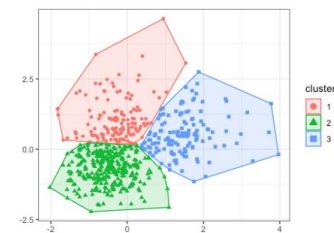
Supervised Learning



Model Evaluation



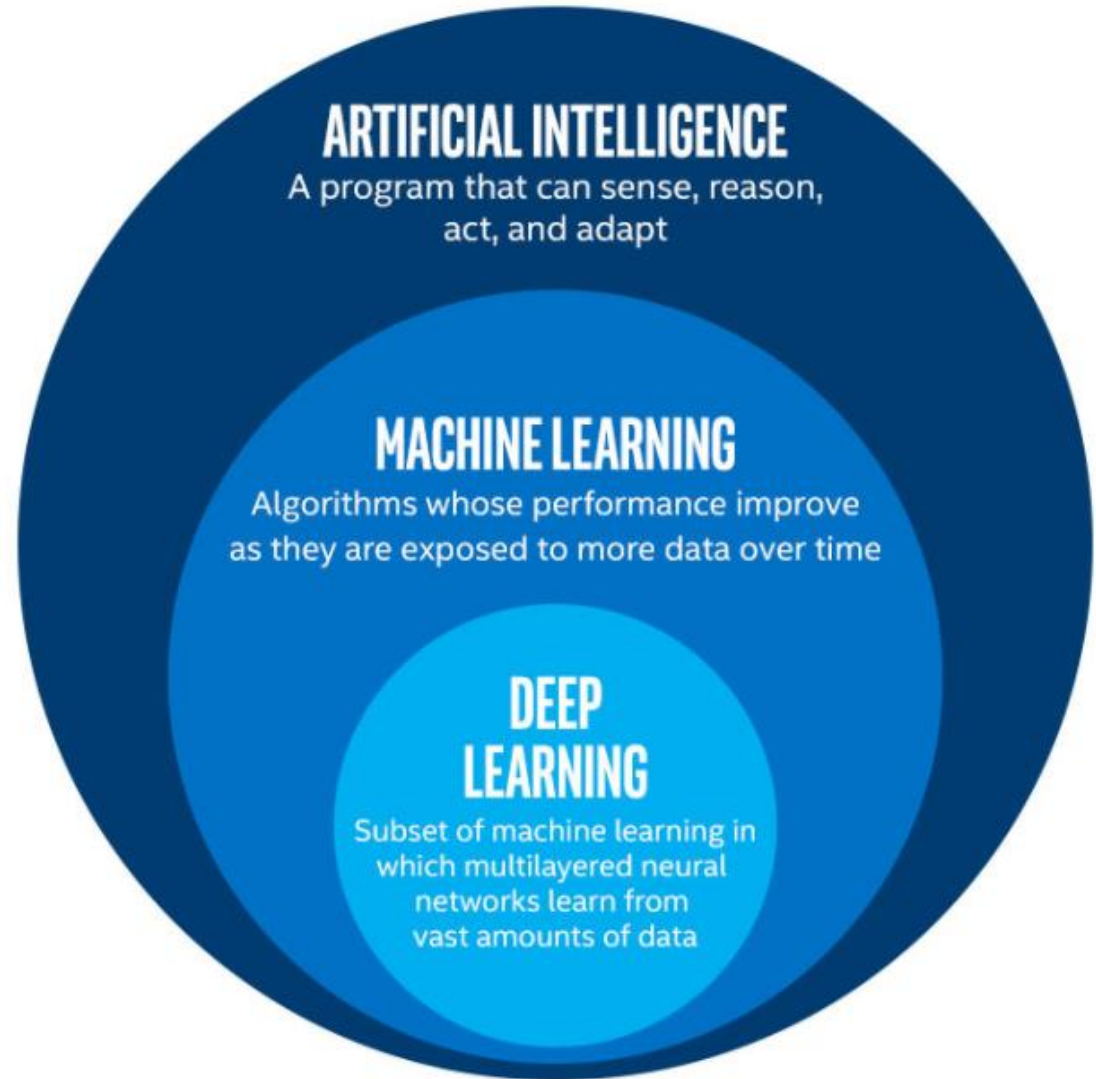
Unsupervised Learning



Tools



Artificial Intelligence



Artificial Intelligence

- **Definition**

- "AI is the capability of a machine to imitate intelligent human behaviour." - John McCarthy, computer scientist and co-founder of AI
- "Artificial intelligence is the branch of computer science that aims to create intelligent machines that work and react like humans." - Techopedia
- "Artificial intelligence is the ability of a computer or machine to mimic or emulate human intelligence in decision-making." - TechTarget
- "Artificial intelligence is the simulation of human intelligence processes by computer systems." - Investopedia



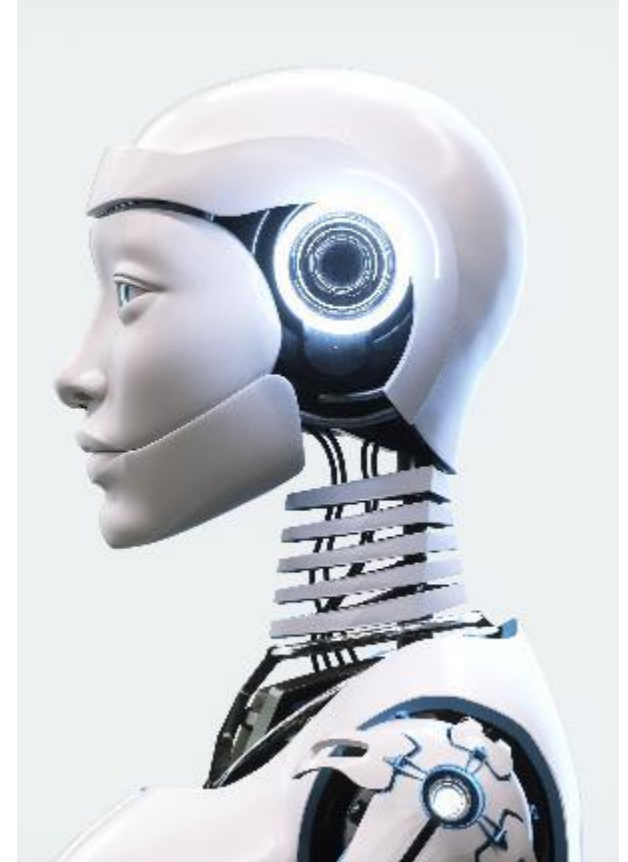
Artificial Intelligence

- **Definition**

- There is no single or fixed definition of AI, but there is common agreement that machines based on AI:

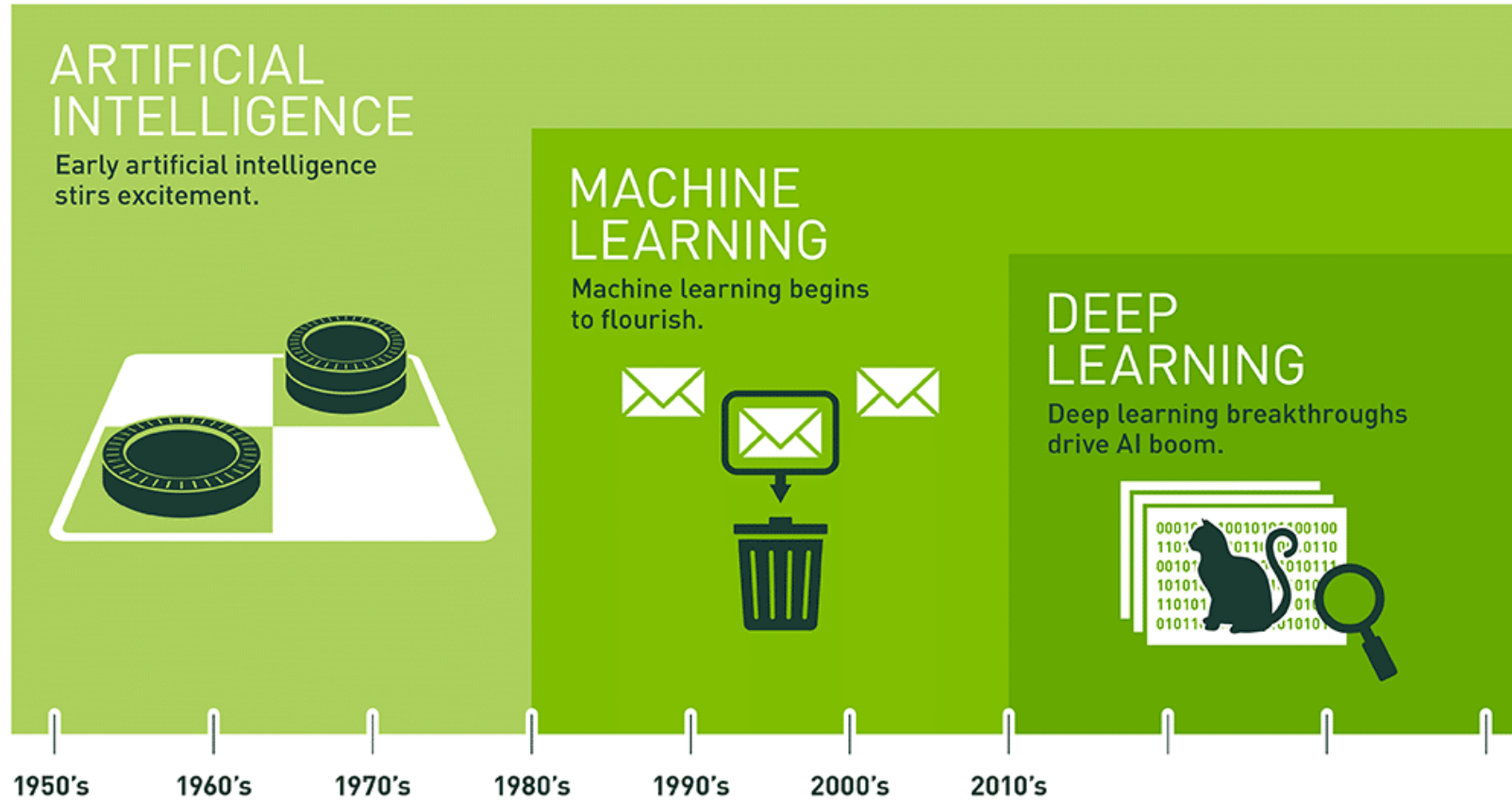
“are potentially capable of imitating or even exceeding human cognitive capacities, including sensing, language interaction, reasoning and analysis, problem solving, and even creativity”

UNESCO (2019). Preliminary Study on the Ethics of Artificial Intelligence. Available at:
<https://unesdoc.unesco.org/ark:/48223/pf0000367823>

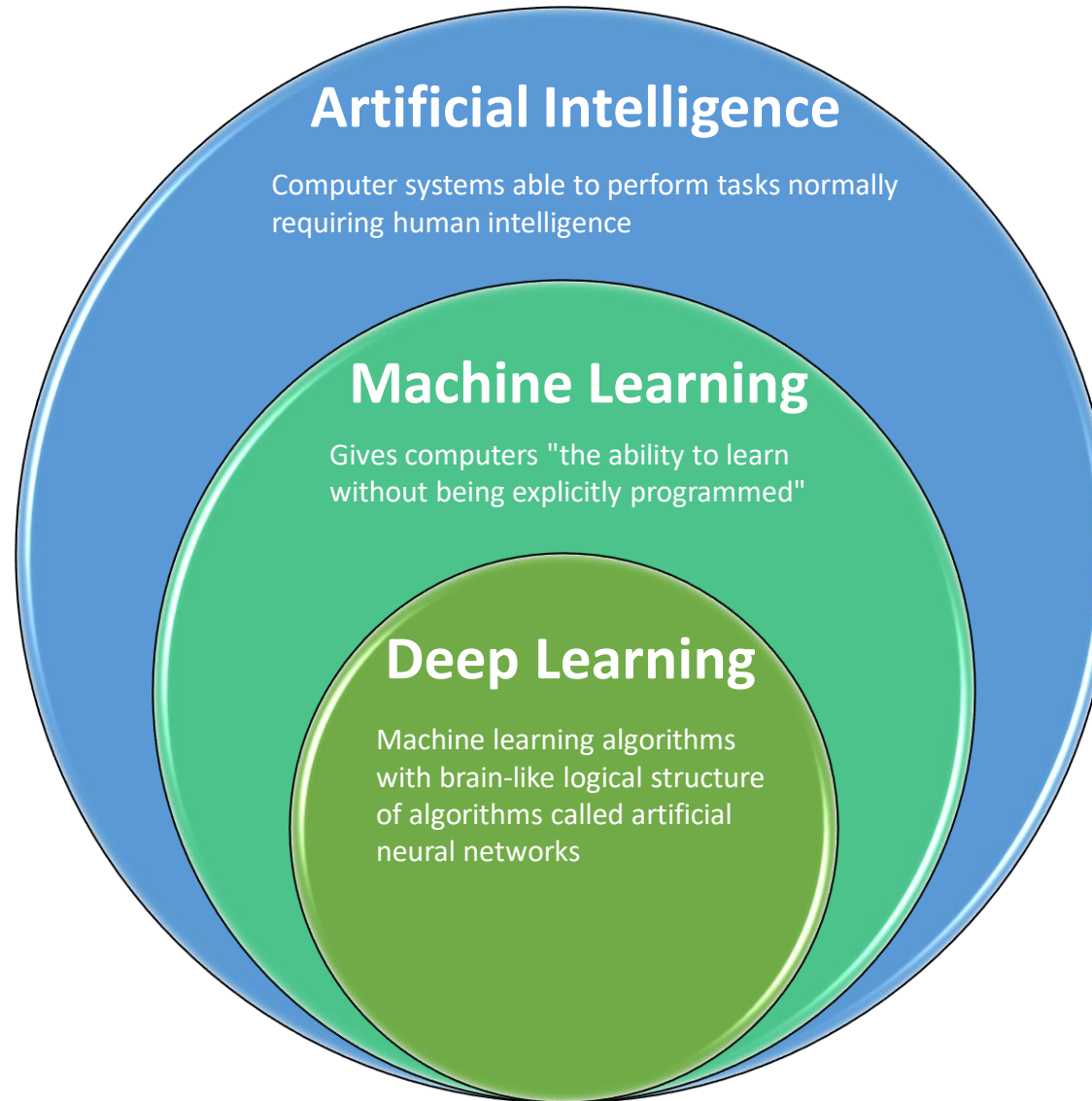


Artificial Intelligence

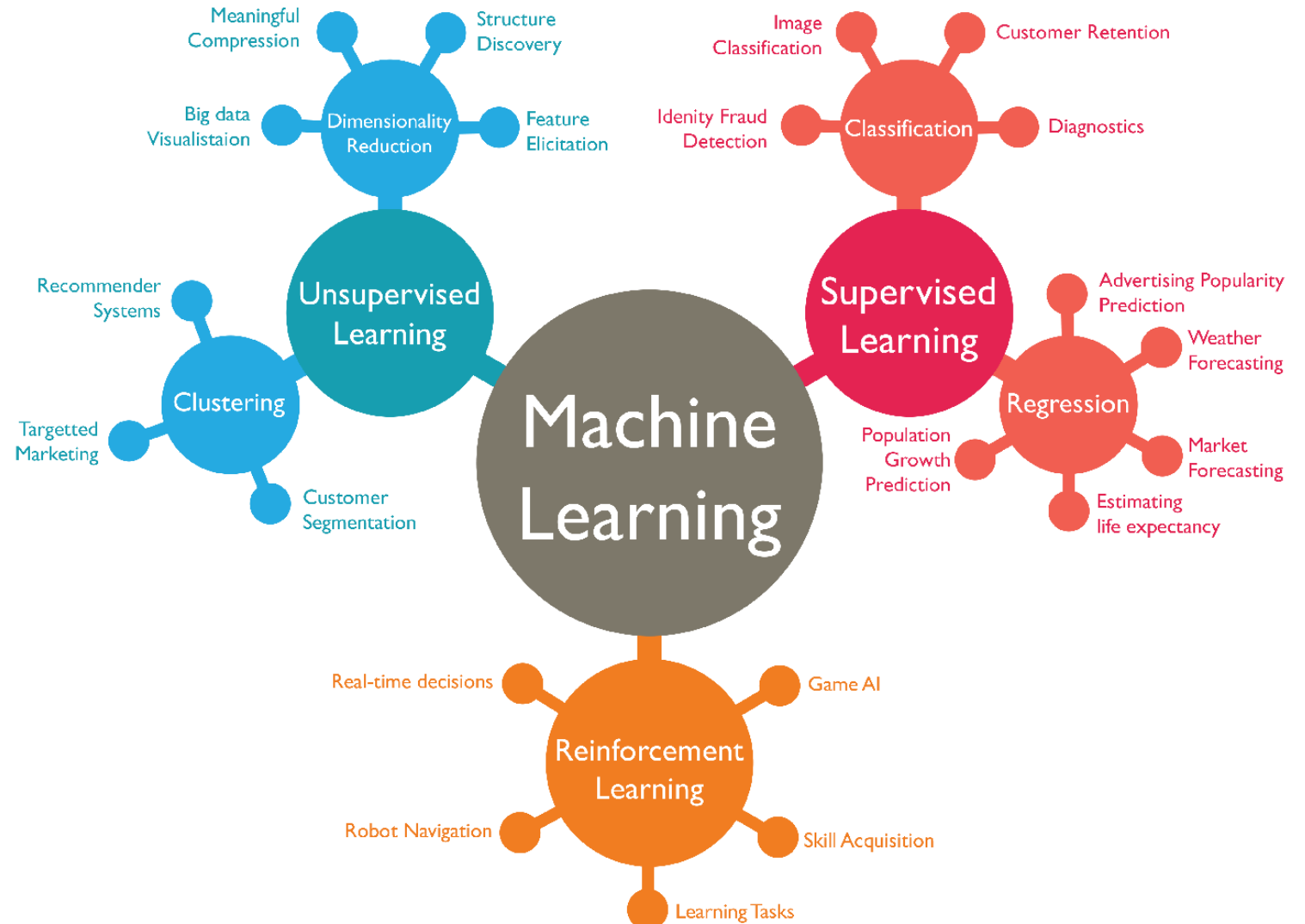
- Timeline



Artificial Intelligence



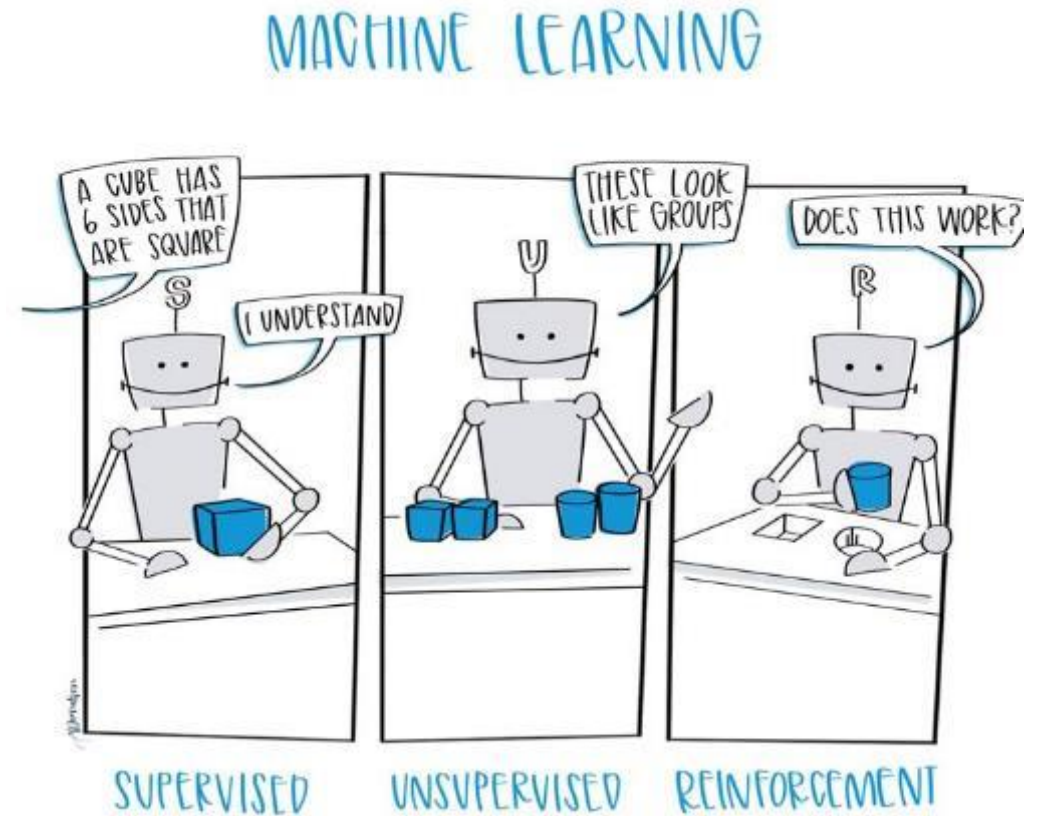
Machine Learning



Machine Learning

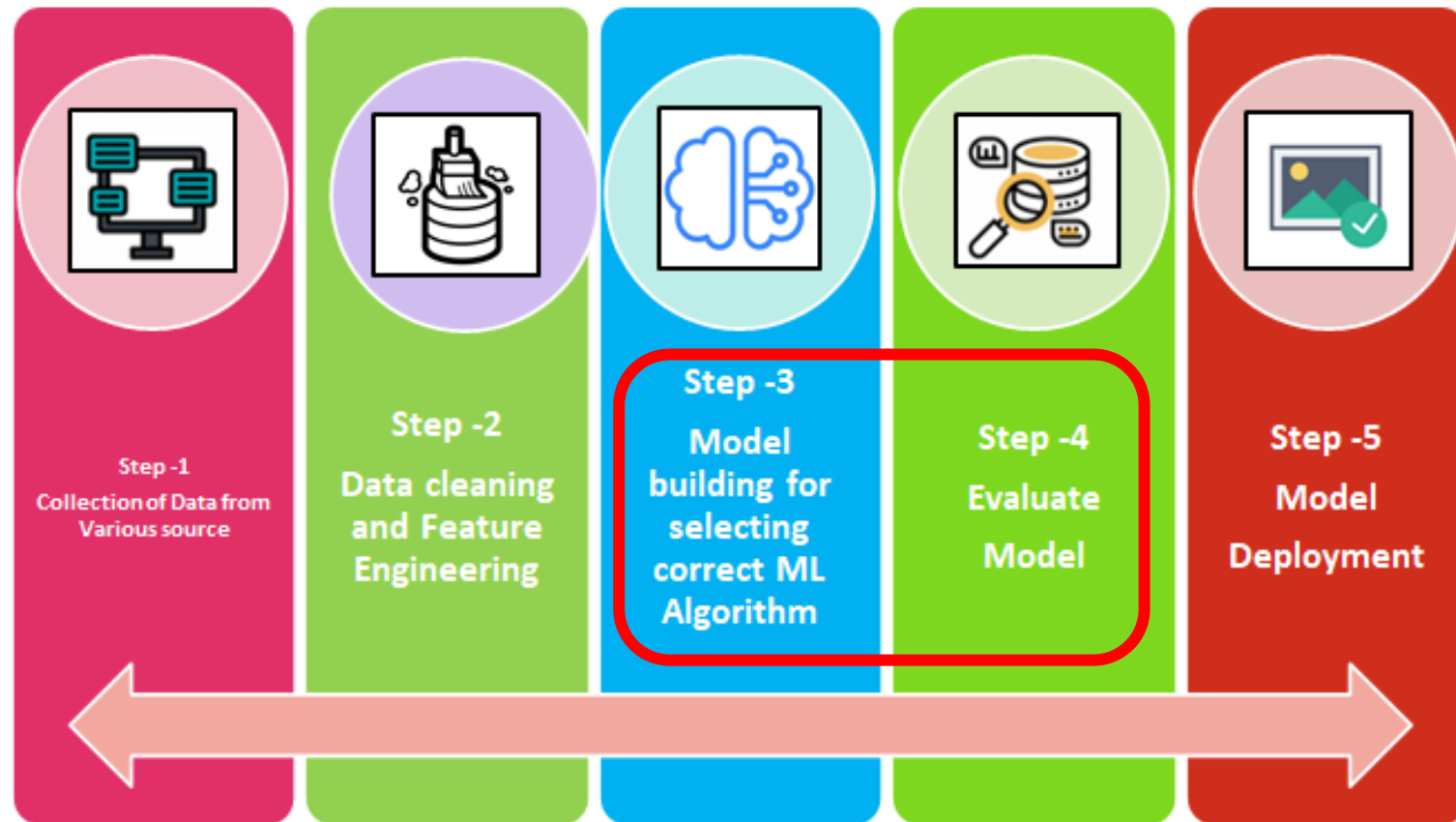
- **Definition**

- "Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed." - Arthur Samuel
- "Machine learning is a subset of artificial intelligence in which algorithms learn from data, identify patterns, and make decisions with minimal human intervention." - Techopedia
- "Machine learning is a method of teaching computers to learn by example, rather than being programmed with specific rules." - NVIDIA



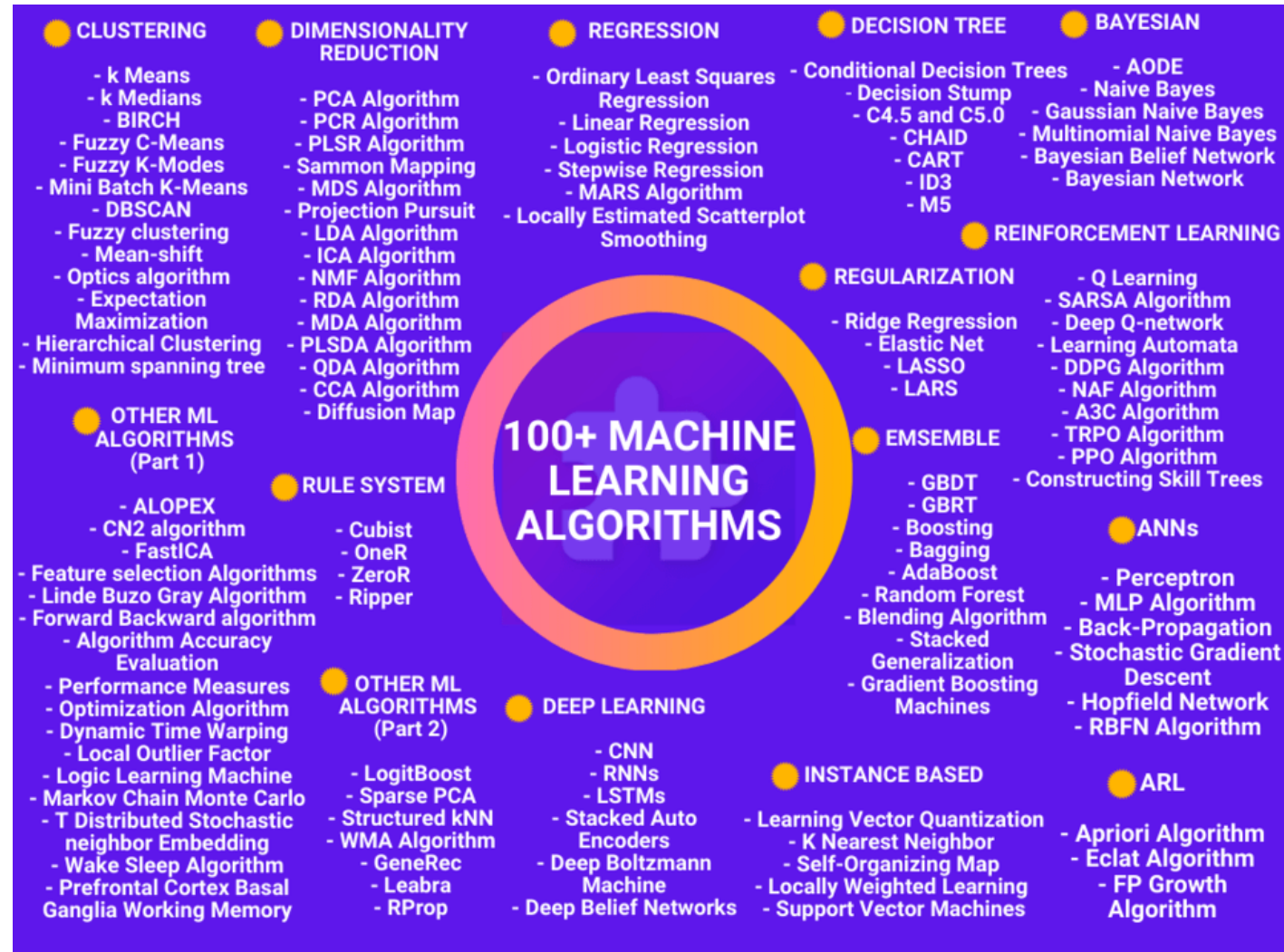
Machine Learning

- Stages



Machine Learning

- Types of algorithms



Machine Learning

- Types of algorithms

Supervised learning

- the training data you feed to the algorithm includes the desired solutions, called labels

Unsupervised learning

- the training data is unlabeled. The system tries to learn without a teacher.

Semisupervised learning

- partially labeled training data, usually a lot of unlabeled data and a little bit of labeled data

Reinforcement Learning

- the learning system can observe the environment, select and perform actions, and get rewards in return (or penalties).

Machine Learning

- Types of algorithms

Supervised learning

- **labelled**
- by **proper guidance** (kind of a teacher)
- explicitly learning model
- **predicts** de outcome/future



Unsupervised learning

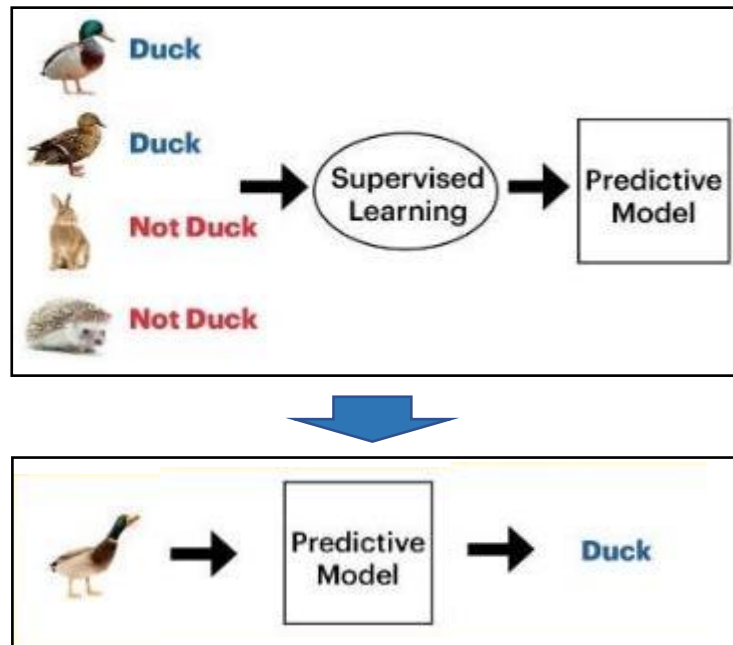
- **NOT labelled**
- no guidance (**self learning**)
- it **identifies the patterns**
- not predict nor find anything specific



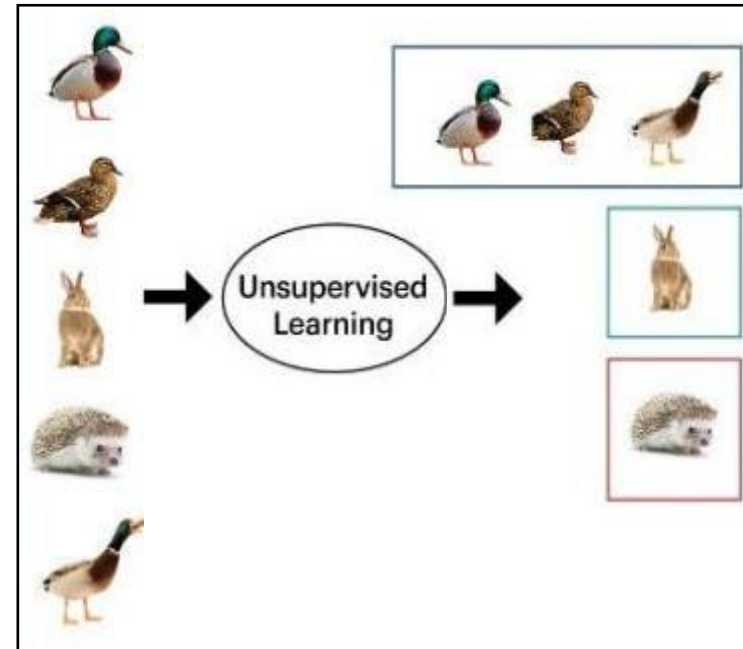
Machine Learning

- Types of algorithms

Supervised learning



Unsupervised learning



Machine Learning

- Types of algorithms

Supervised learning

X_1	X_2	X_3	X_p	Y
				Duck
				Duck
				Not Duck
				Duck

Target

Unsupervised learning

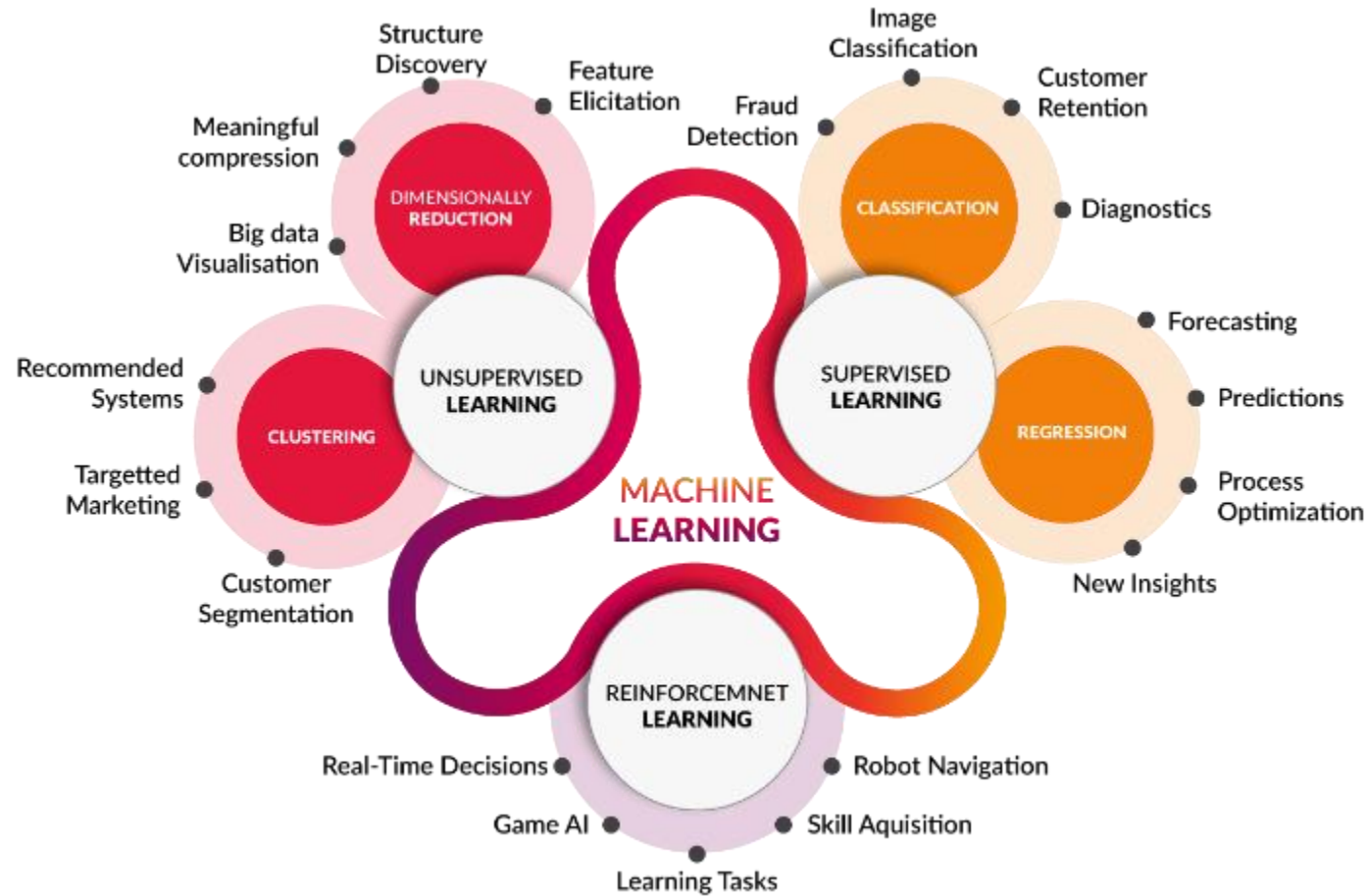
X_1	X_2	X_3	X_p	Y

No
Target

<https://www.sharpsightlabs.com/blog/supervised-vs-unsupervised-learning/>

Machine Learning

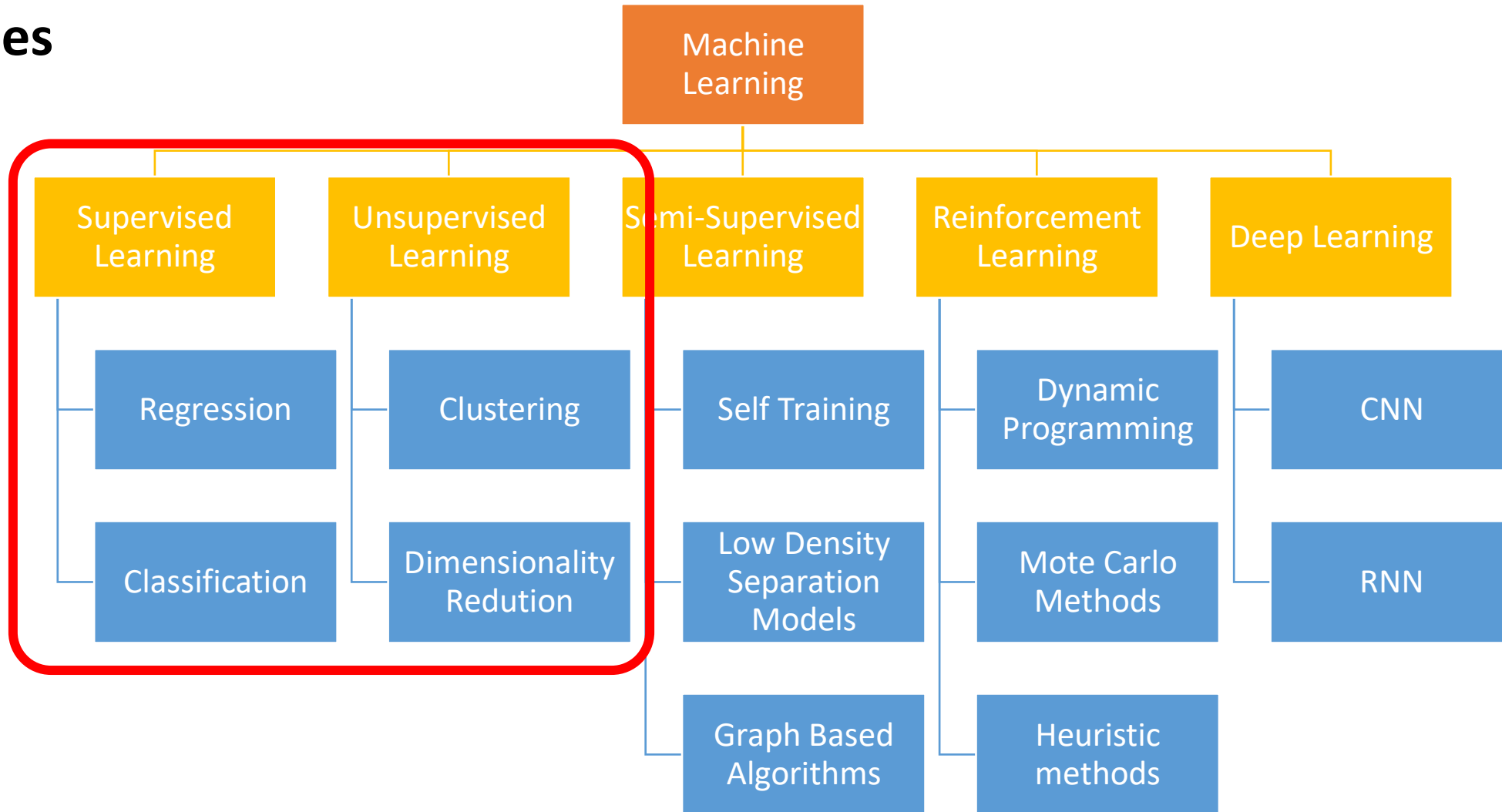
- Types



<https://towardsdatascience.com/coding-deep-learning-for-beginners-types-of-machine-learning-b9e651e1ed9d>

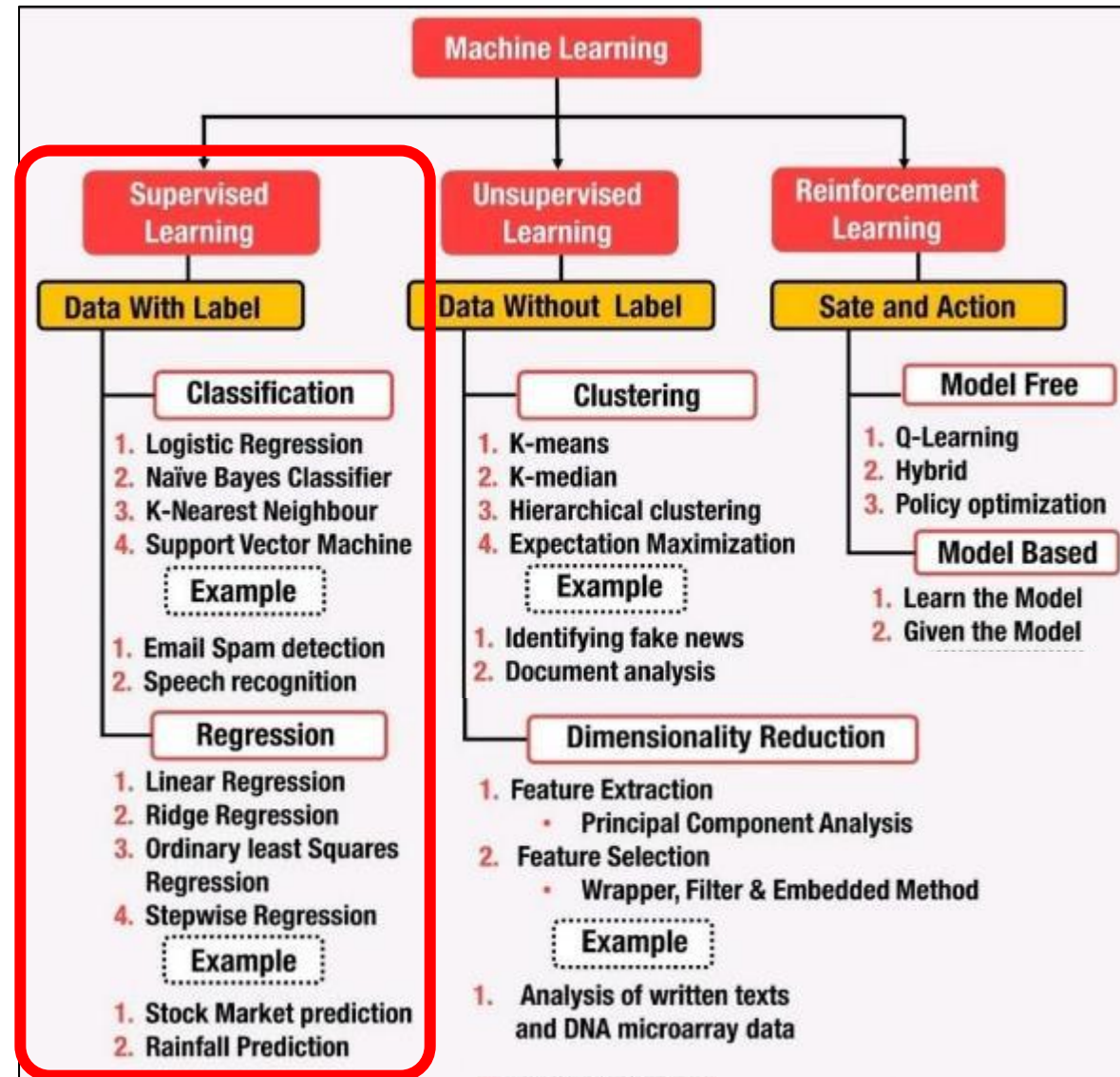
Machine Learning

- Types



Machine Learning

- Types



Machine Learning

- Types

Go to mentimeter.com and use the code 3496 0557

Which of the following ML algorithms uses labels?

0

0

0

Supervised

Unsupervised

Reinforcement Learning

Content

Your question ⓘ
Which of the following ML algorithms uses

Add longer description

Options ⓘ
Supervised X
Unsupervised X
Reinforcement Learning X
+ Add

Extras
Show correct answer(s) ⓘ ☐
Show results in percentage ⓘ ☐
Let participants submit multiple options ☐
Remove question

LV

Account

Content

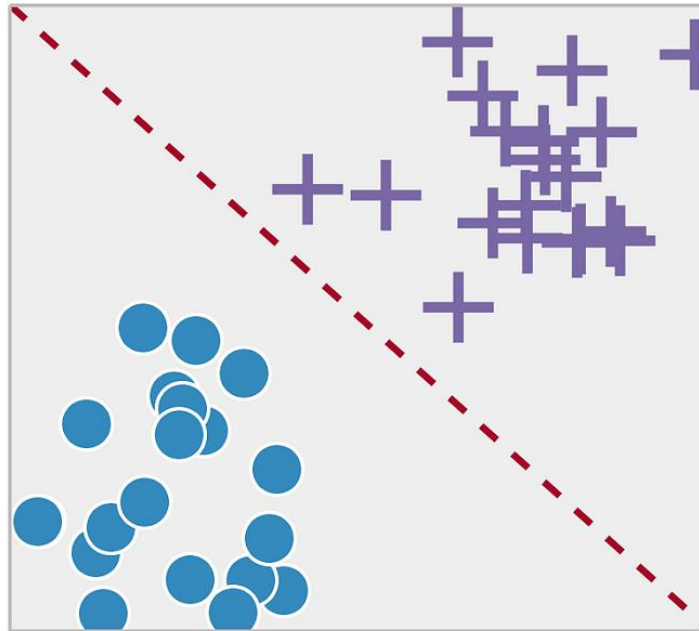
Design

Settings

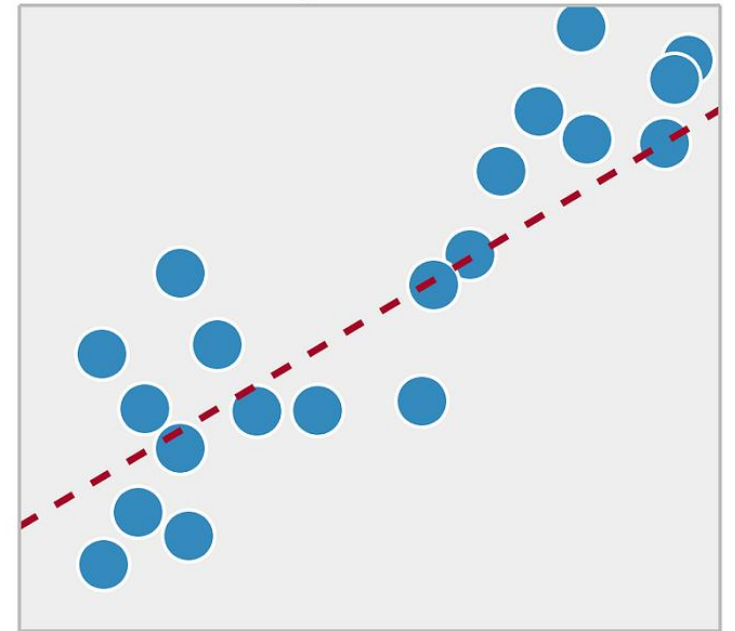
Got feedback?

Supervised Learning

Classification

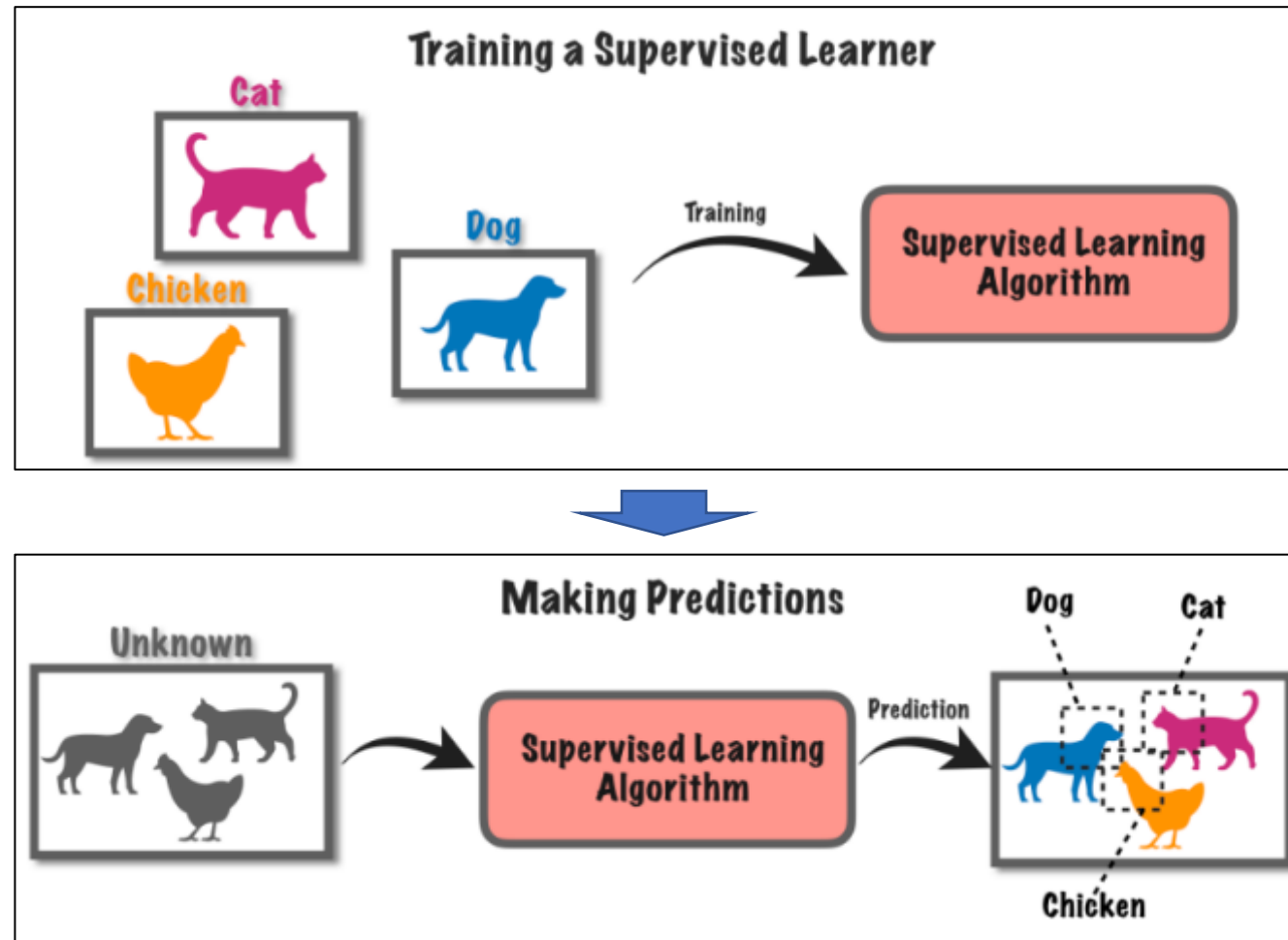


Regression



Machine Learning

- Supervised Learning
 - Definition



Machine Learning

- Supervised Learning

Attribute

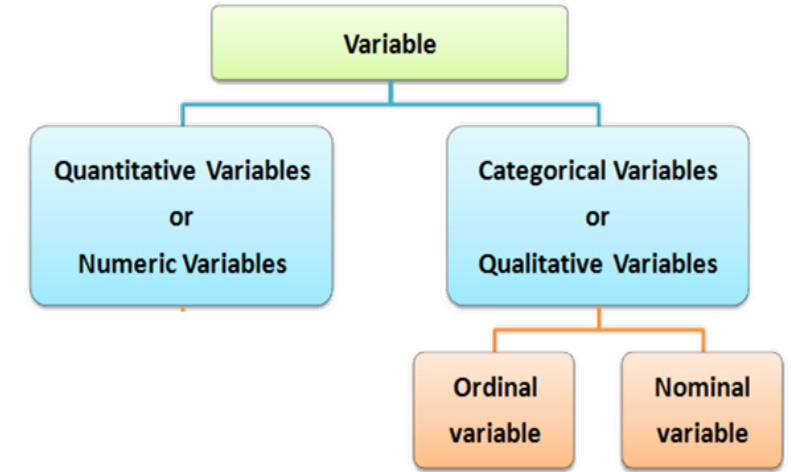
Feature

Instance (sample, observation)

Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	High	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A

X: independent variables

Y: dependent variable / target variable



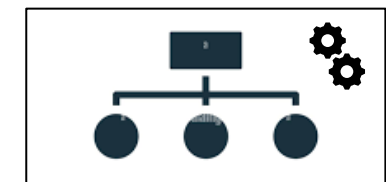
Machine Learning

- Supervised Learning

Target /
categorical
variable

Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	High	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A

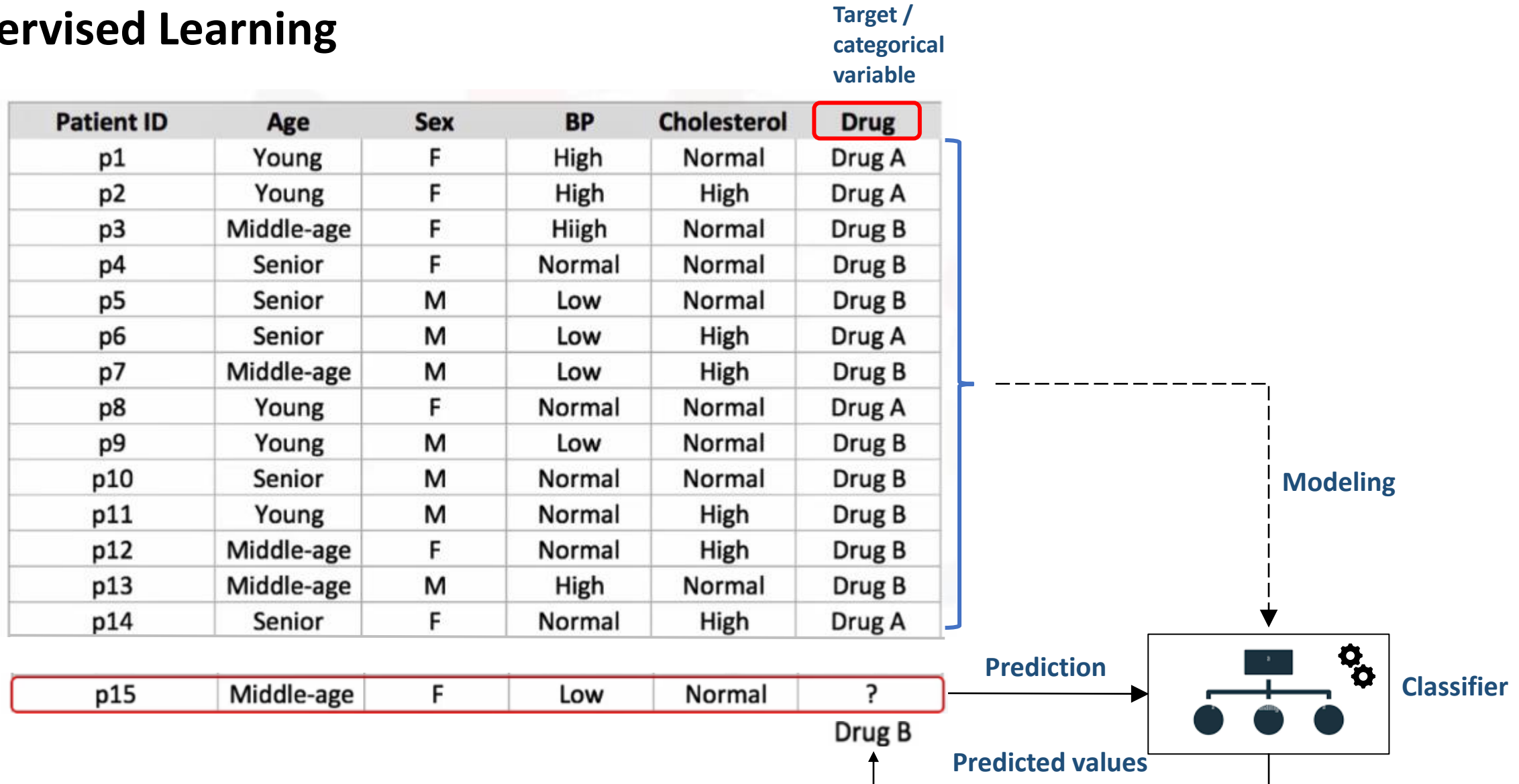
Modeling



Classifier

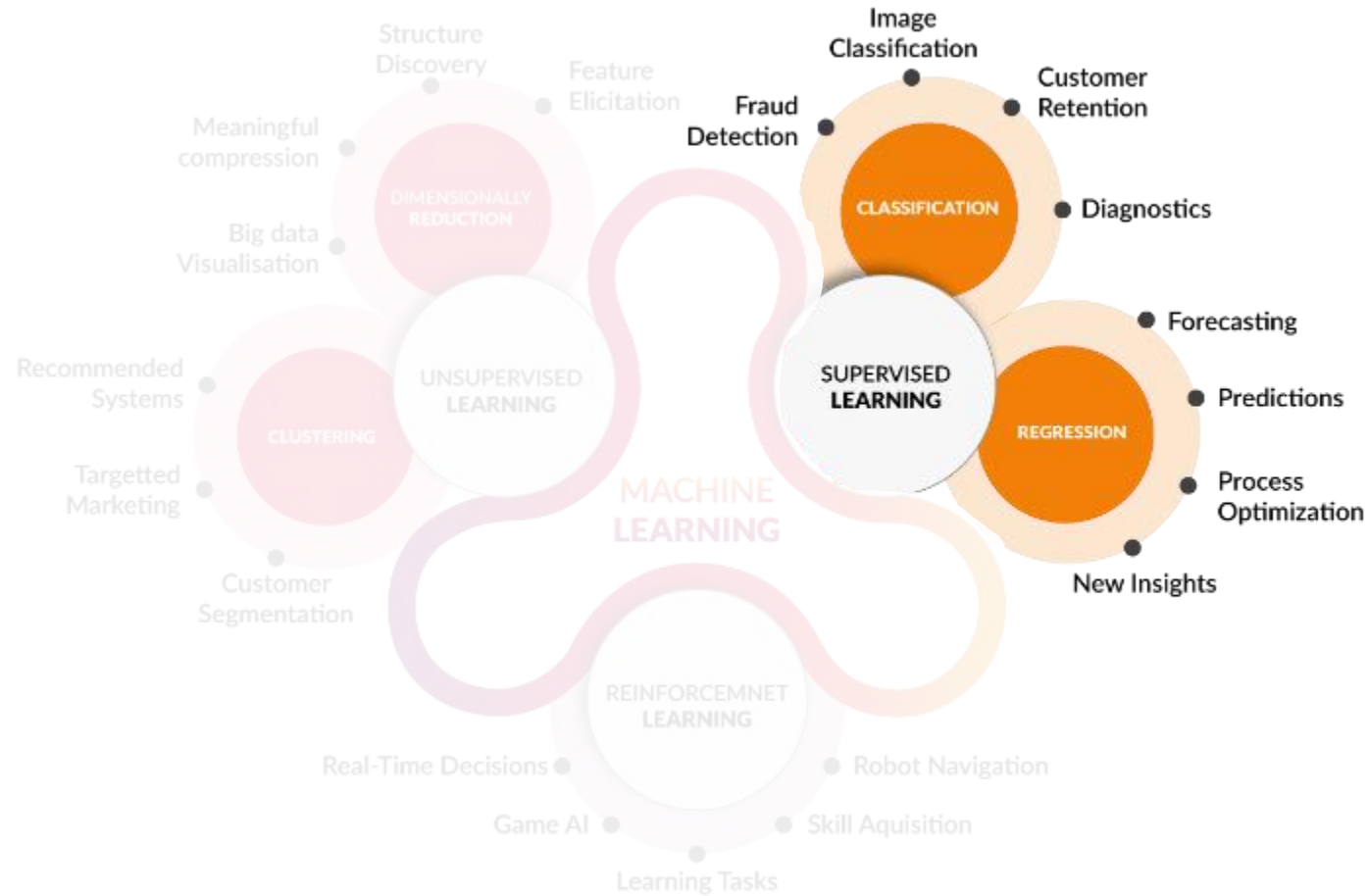
Machine Learning

- Supervised Learning



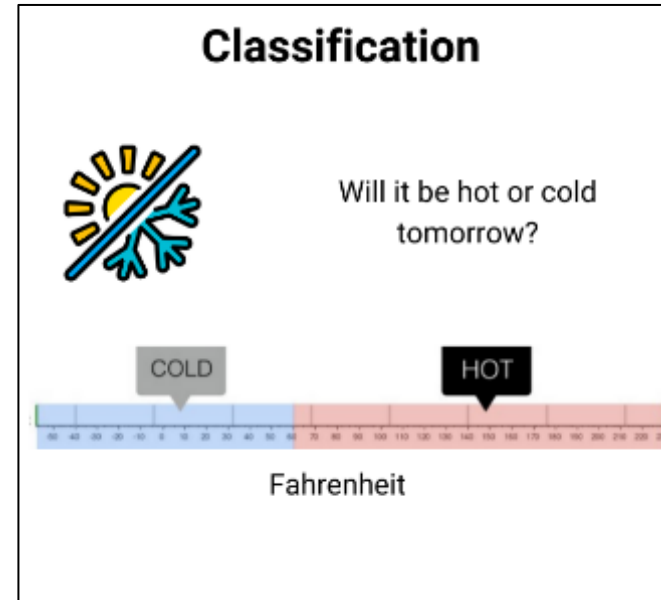
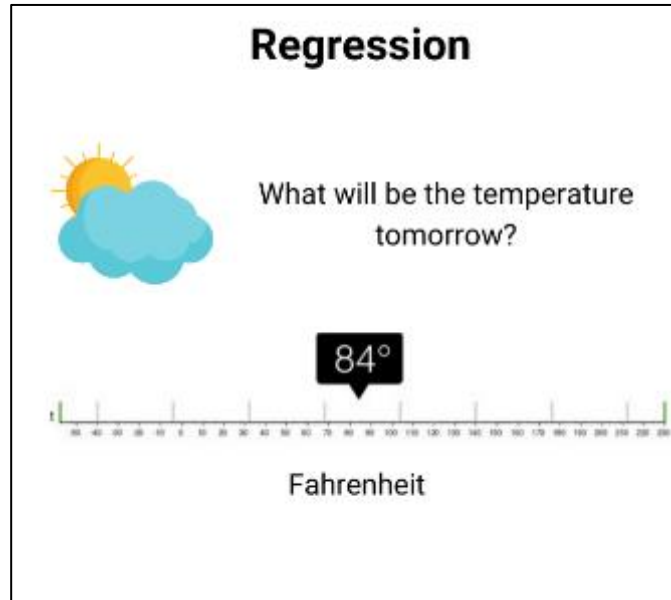
Machine Learning

- Supervised Learning
 - Types



Machine Learning

- Supervised Learning
 - Types

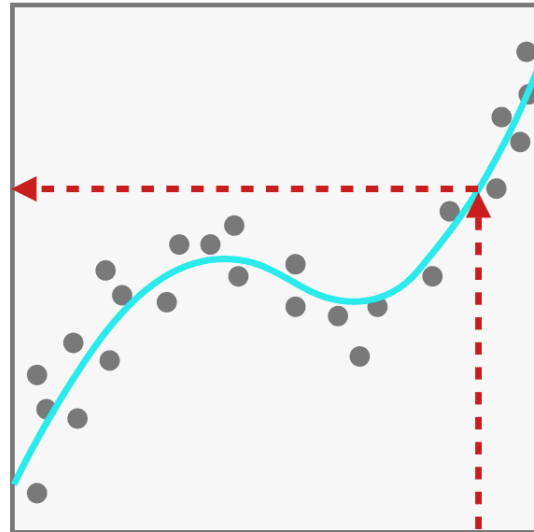


<https://www.enjoyalgorithms.com/blogs/classification-and-regression-in-machine-learning>

Machine Learning

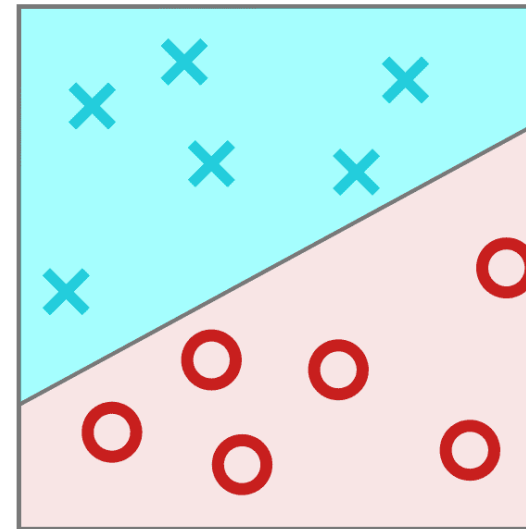
- Supervised Learning
 - Types

Regression predicts a numeric value



Here, the fitted line provides a predicted output, if we give it an input

Classification Groups observations into "classes"



Here, the line classifies the observations into X's and O's

<https://www.sharpsightlabs.com/blog/regression-vs-classification/>

Machine Learning

- Supervised Learning

- Types

Regression

- process of predicting a **continuous, numerical value** for input data sample.
- Example usages: assessing the house price, forecasting grocery store food demand, temperature forecasting.

X ₁	X ₂	X ₃	X _p	Y
				5.2
				1.3
				23.0
				7.4

Numeric
Target

Classification

- process of assigning **category** to input data sample.
- Example usages: predicting whether a person is ill or not, detecting fraudulent transactions, face classifier.

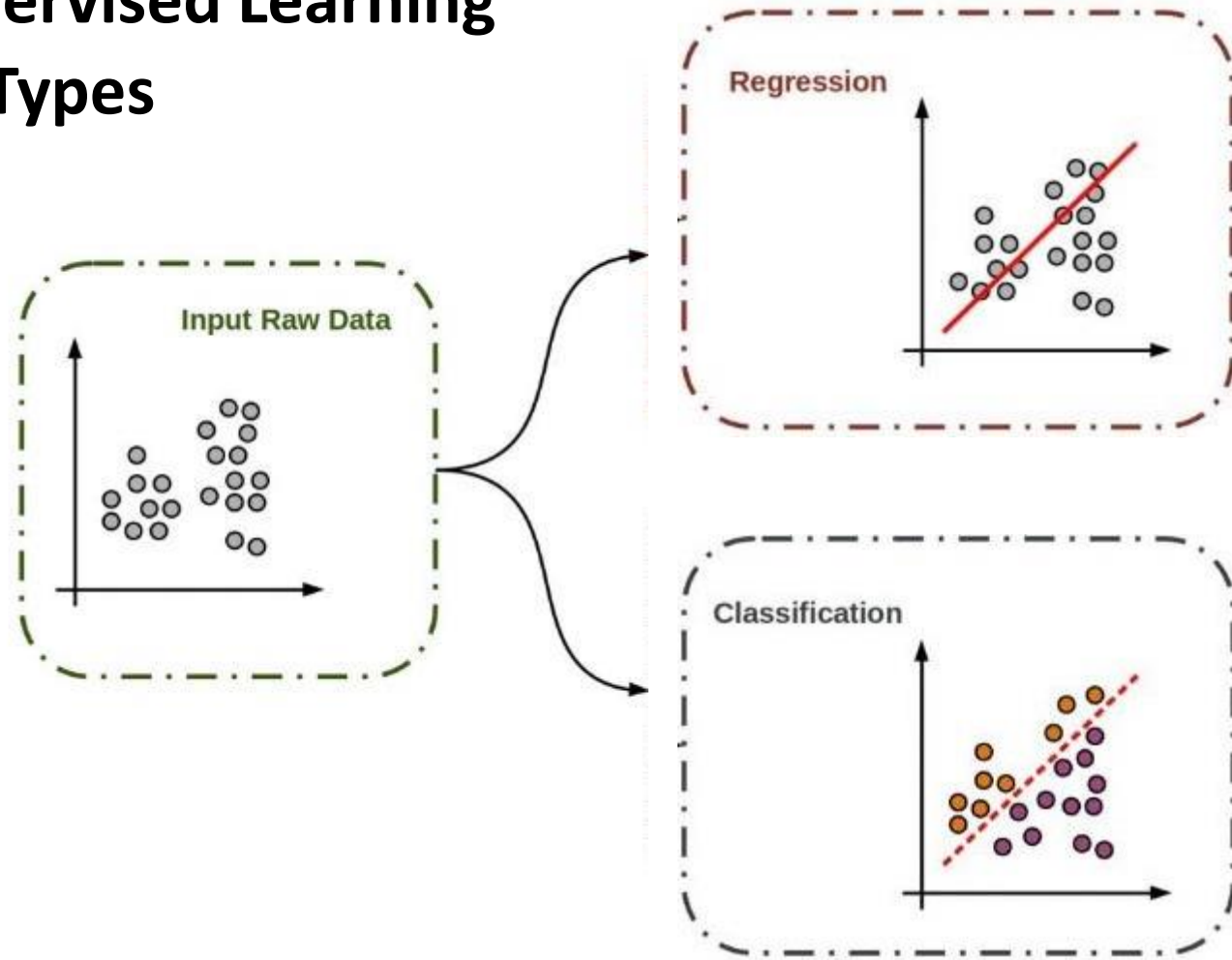
X ₁	X ₂	X ₃	X _p	Y
				cat
				dog
				cat
				cat

Categorical
"Labels"

<https://www.sharpsightlabs.com/blog/regression-vs-classification/>

Machine Learning

- Supervised Learning
 - Types



Dependent
variable / Target

Age	Income	Loan Amount
21	20000	0
37	55000	150000
29	35000	120000
23	17000	550000
34	70000	250000
47	84000	0
25	30000	90000

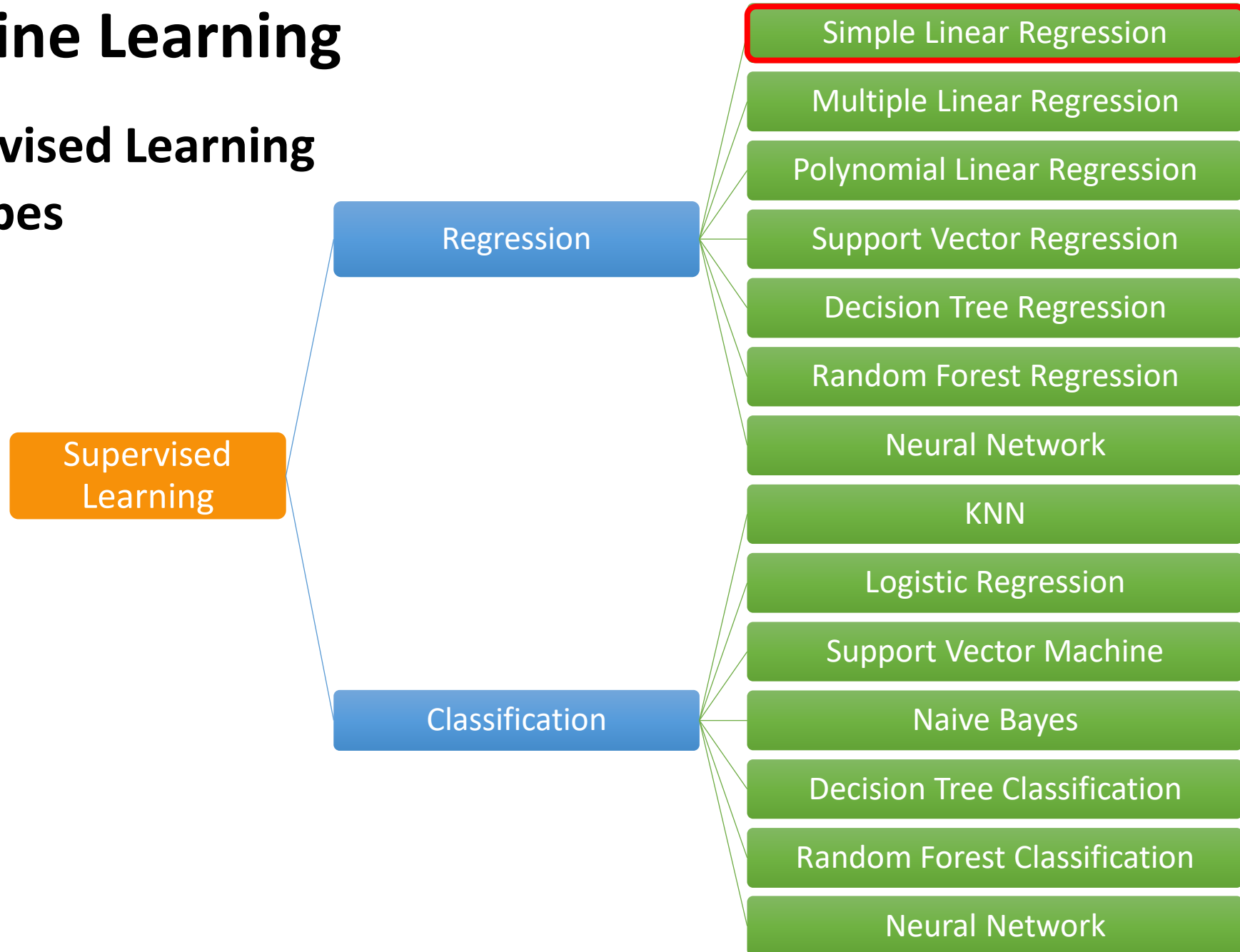
Continuous
variable

Age	Income	Loan Status
21	20000	Rejected
37	55000	Approved
29	35000	Approved
23	17000	Rejected
34	70000	Approved
47	84000	Rejected
25	30000	Approved

Categorical
(nominal)
variable

Machine Learning

- Supervised Learning
 - Types



Machine Learning

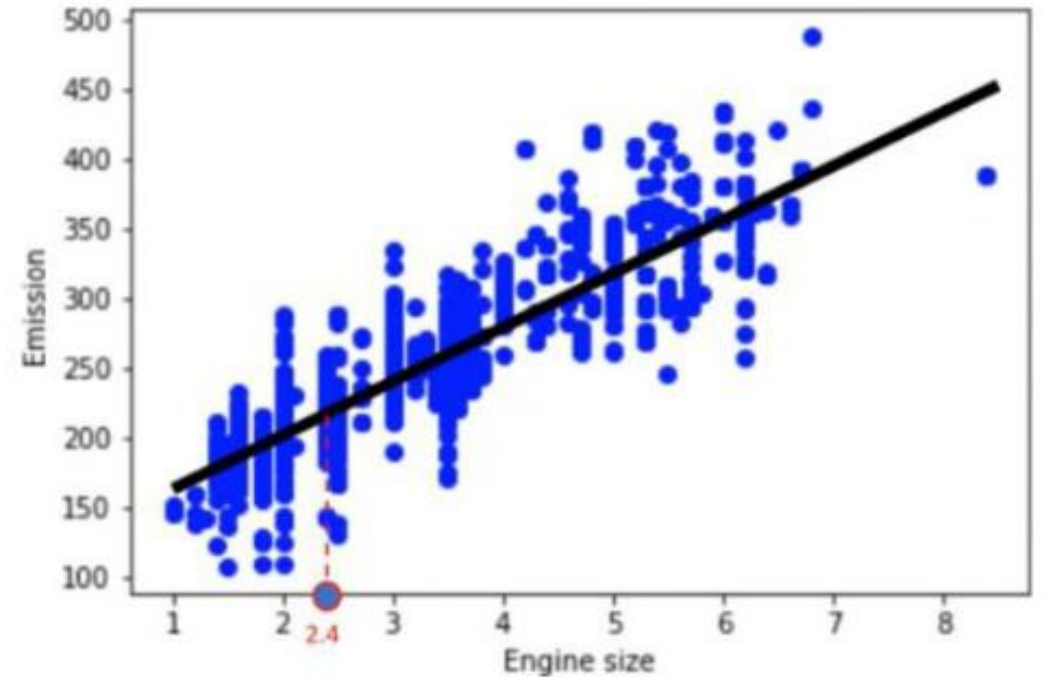
- Supervised Learning
 - Linear Regression

X: independent variables

Y: dependent variable

	ENGINE SIZE	CYLINDERS	FUEL CONSUMPTION_COMB	CO2 EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	?

continuous



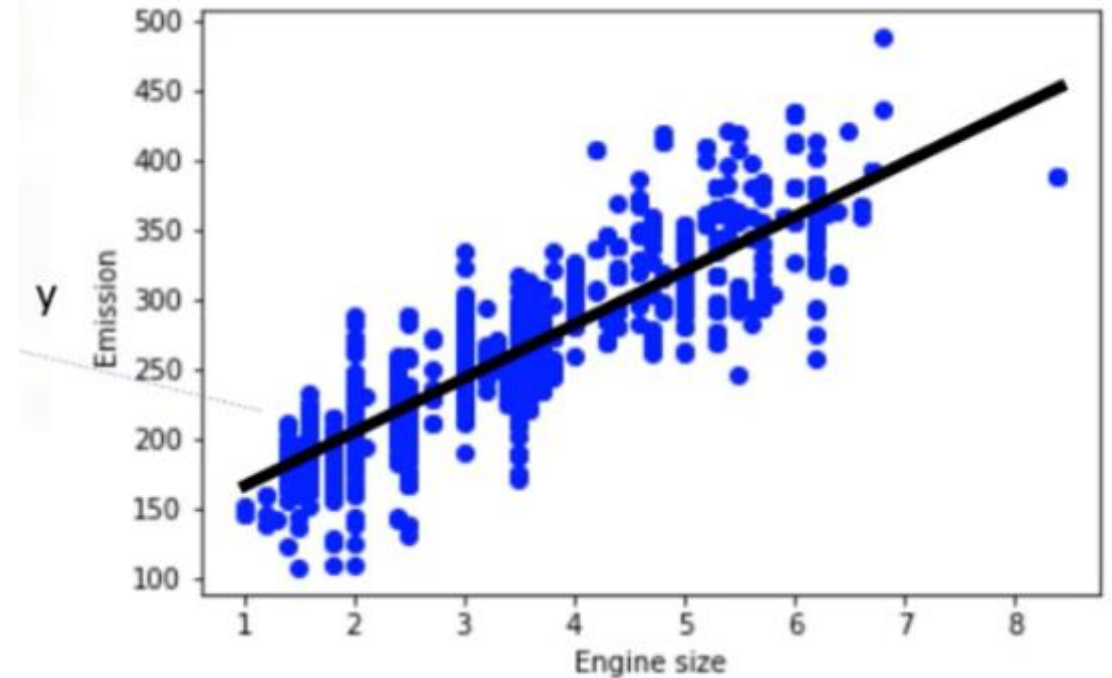
Machine Learning

- Supervised Learning
 - Regression

$$\hat{y} = \theta_0 + \theta_1 x_1$$

θ_0 - slope or gradient

θ_1 - value when $x=0$ (i.e. intercept on the y-axis)



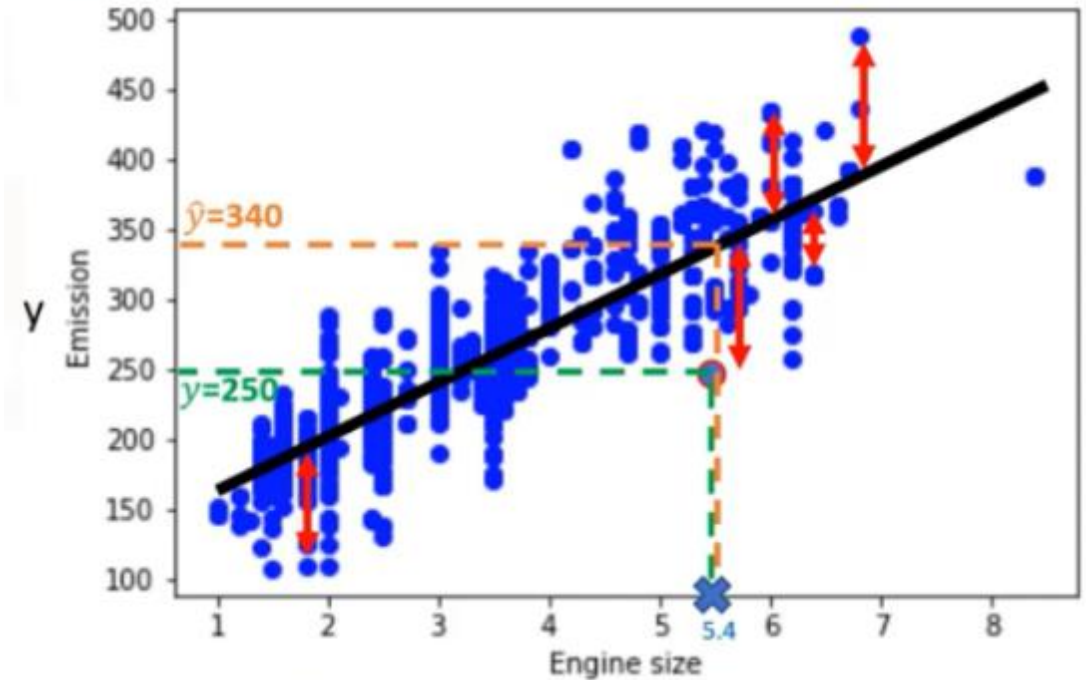
Machine Learning

- Supervised Learning
 - Regression

$$\hat{y} = \theta_0 + \theta_1 x_1$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Mean Squared Error (MSE) - it is the mean of all residual errors and shows how well the line fits with the whole dataset.



Machine Learning

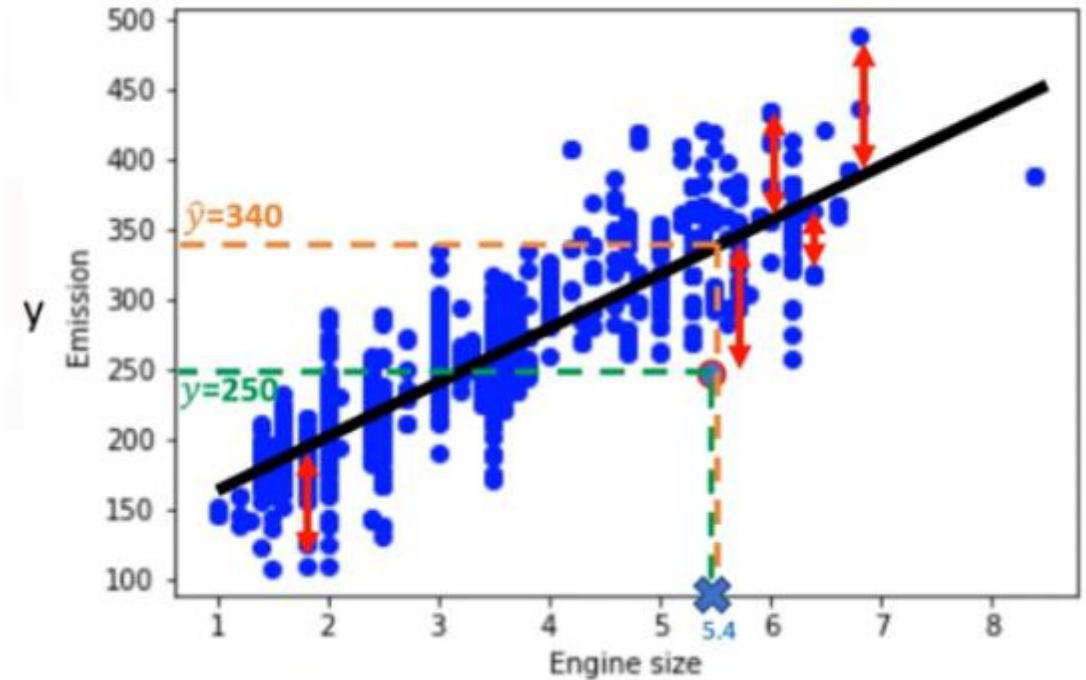
- Supervised Learning
 - Regression

$$\hat{y} = \theta_0 + \theta_1 x_1$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

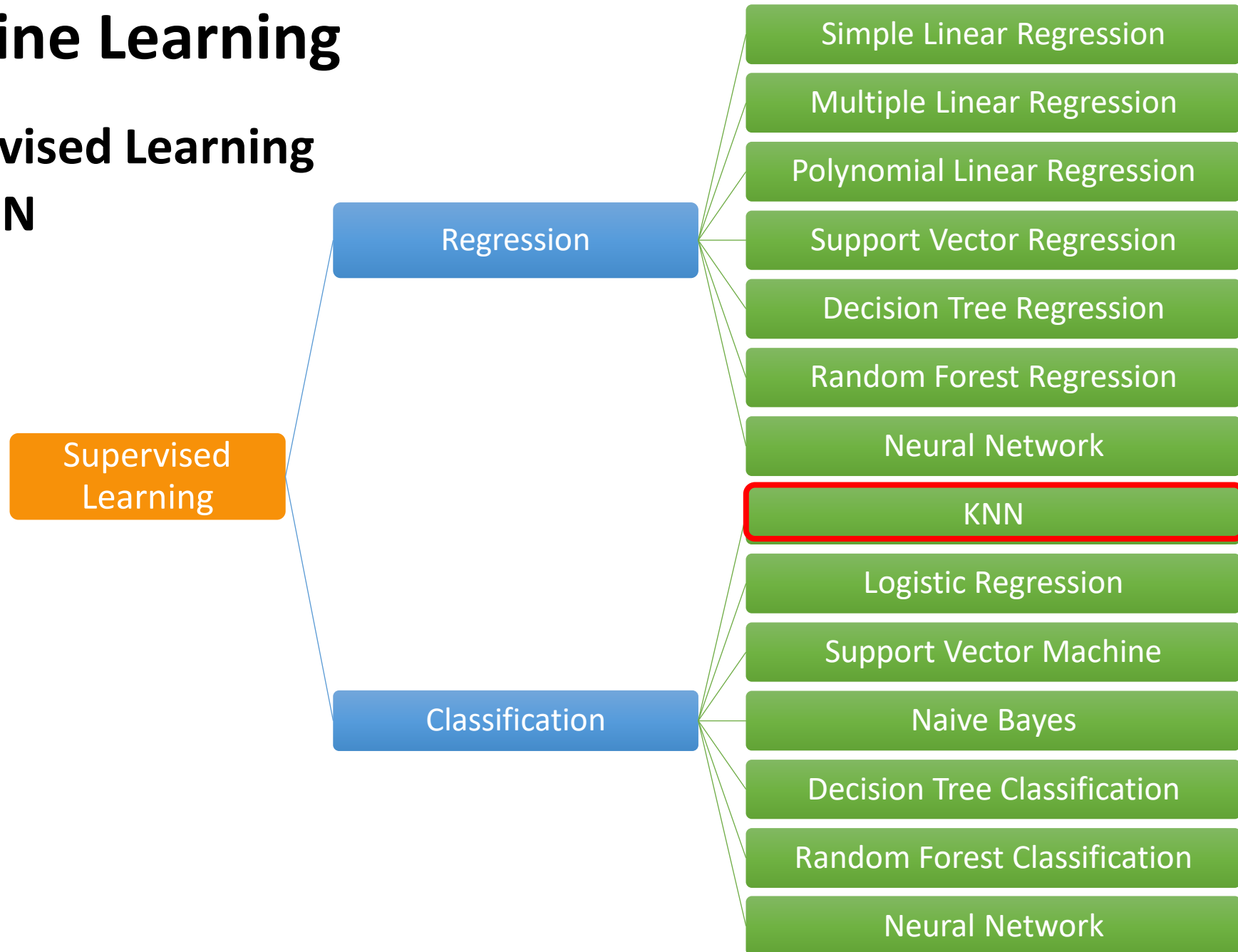
Objective:

- to find a line where the mean of all these errors (MSE) is minimized.
- find the best parameters, **θ_0** and **θ_1** .



Machine Learning

- Supervised Learning
 - KNN



Machine Learning

- Supervised Learning
 - KNN

	region	age	marital	address	income	ed	employ	retire	gender	reside	custcat
0	2	44	1	9	64	4	5	0	0	2	1
1	3	33	1	7	136	5	5	0	0	6	4
2	3	52	1	24	116	1	29	0	1	2	3
3	2	33	0	12	33	2	0	0	1	1	1
4	2	30	1	9	30	1	2	0	0	4	3
5	2	39	0	17	78	2	16	0	1	1	3
6	3	22	1	2	19	2	4	0	1	5	2
7	2	35	0	5	76	2	10	0	0	3	4
8	3	50	1	7	166	4	31	0	0	5	?

X: independente variable

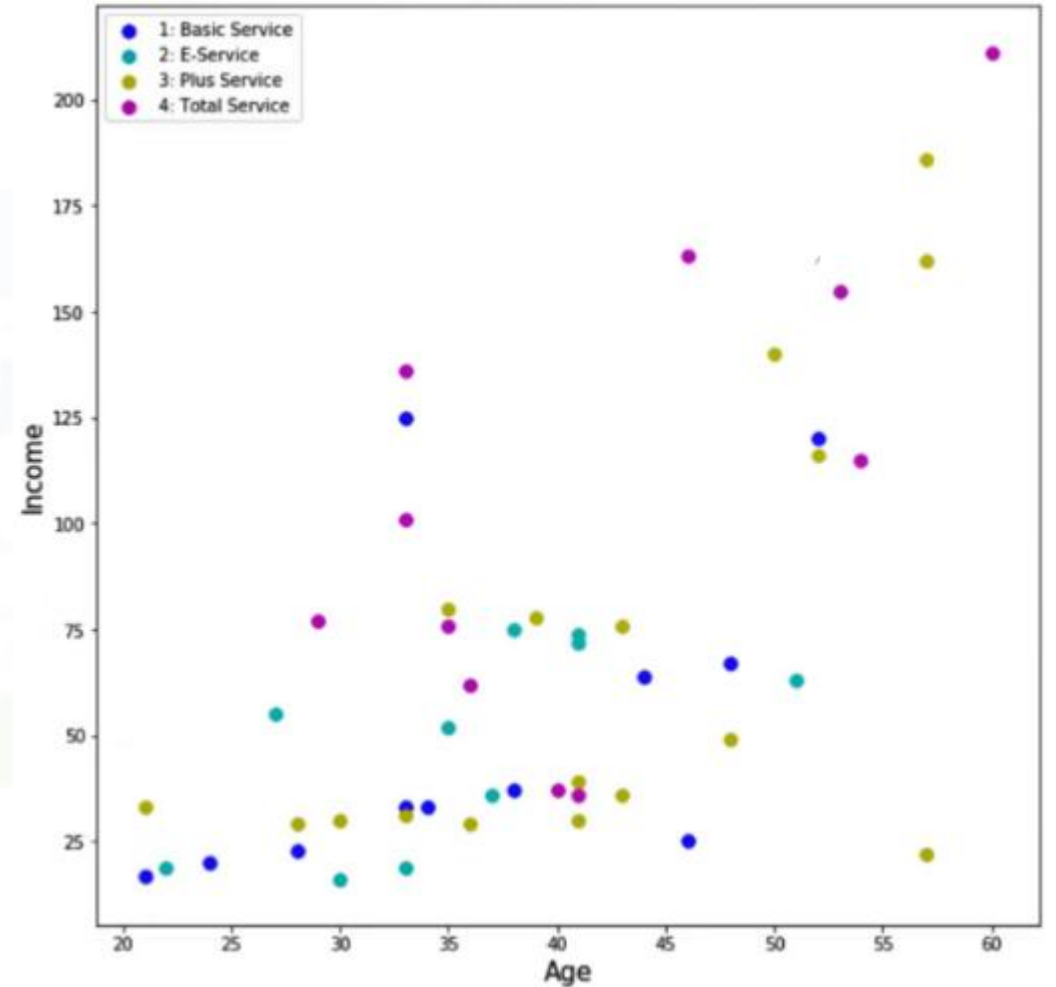
Y: dependente variable

Value	Label
1	Basic Service
2	E-Service
3	Plus Service
4	Total Service

Machine Learning

- Supervised Learning
 - KNN

	region	age	marital	address	income	ed	employ	retire	gender	reside	custcat
0	2	44	1	9	64	4	5	0	0	2	1
1	3	33	1	7	136	5	5	0	0	6	4
2	3	52	1	24	116	1	29	0	1	2	3
3	2	33	0	12	33	2	0	0	1	1	1
4	2	30	1	9	30	1	2	0	0	4	3
5	2	39	0	17	78	2	16	0	1	1	3
6	3	22	1	2	19	2	4	0	1	5	2
7	2	35	0	5	76	2	10	0	0	3	4
8	3	50	1	7	166	4	31	0	0	5	?



- **Supervised Learning**
 - **KNN**

1-NN $\rightarrow$ 4: Total Service

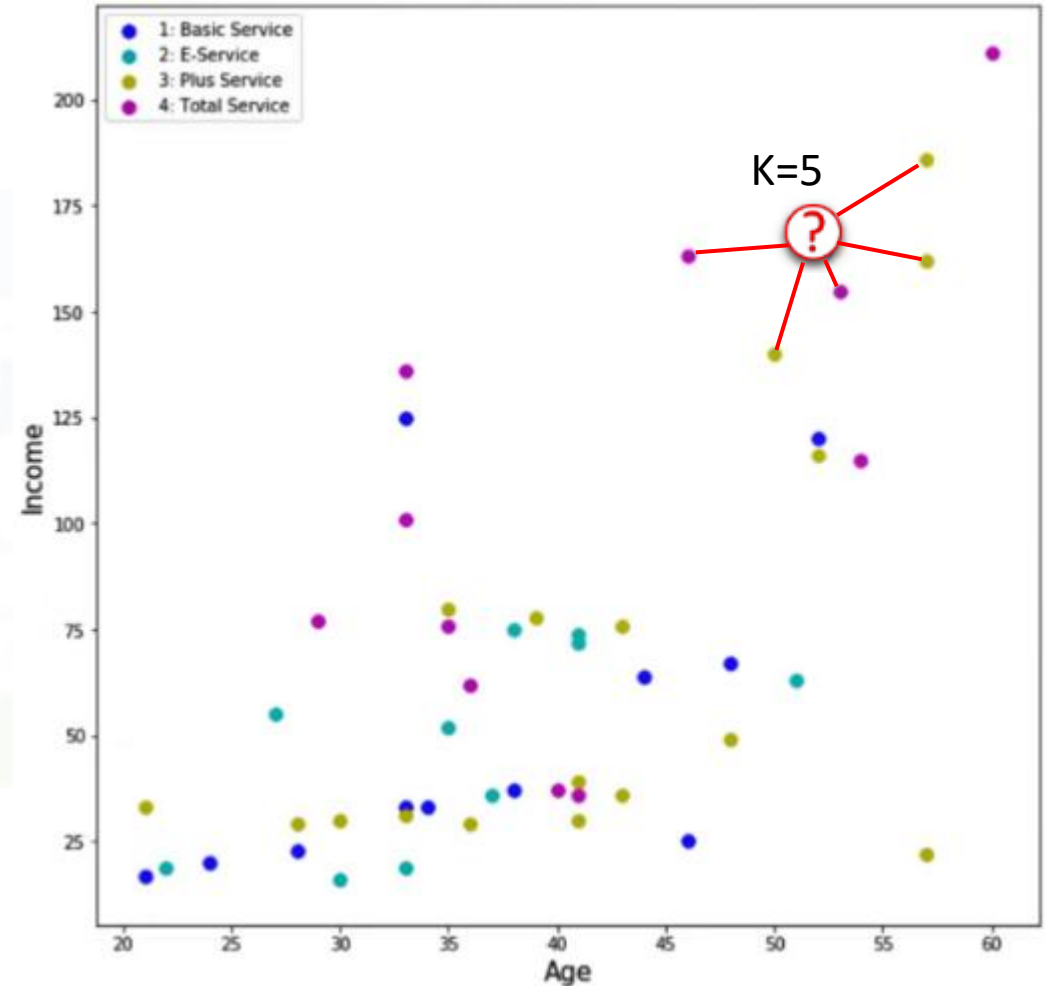


Machine Learning

- Supervised Learning
 - KNN

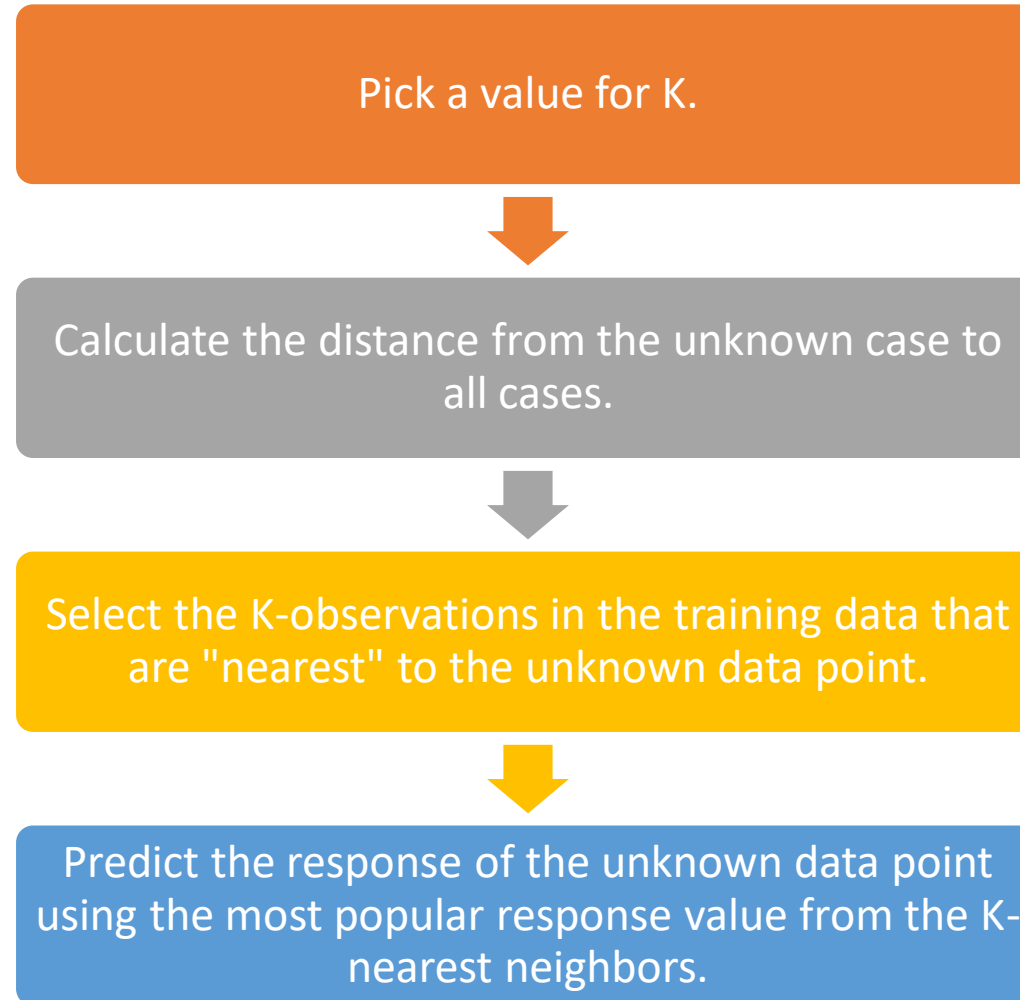
	region	age	marital	address	income	ed	employ	retire	gender	reside	custcat
0	2	44	1	9	64	4	5	0	0	2	1
1	3	33	1	7	136	5	5	0	0	6	4
2	3	52	1	24	116	1	29	0	1	2	3
3	2	33	0	12	33	2	0	0	1	1	1
4	2	30	1	9	30	1	2	0	0	4	3
5	2	39	0	17	78	2	16	0	1	1	3
6	3	22	1	2	19	2	4	0	1	5	2
7	2	35	0	5	76	2	10	0	0	3	4
8	3	50	1	7	166	4	31	0	0	5	?

5-NN → 3: Plus Service



Machine Learning

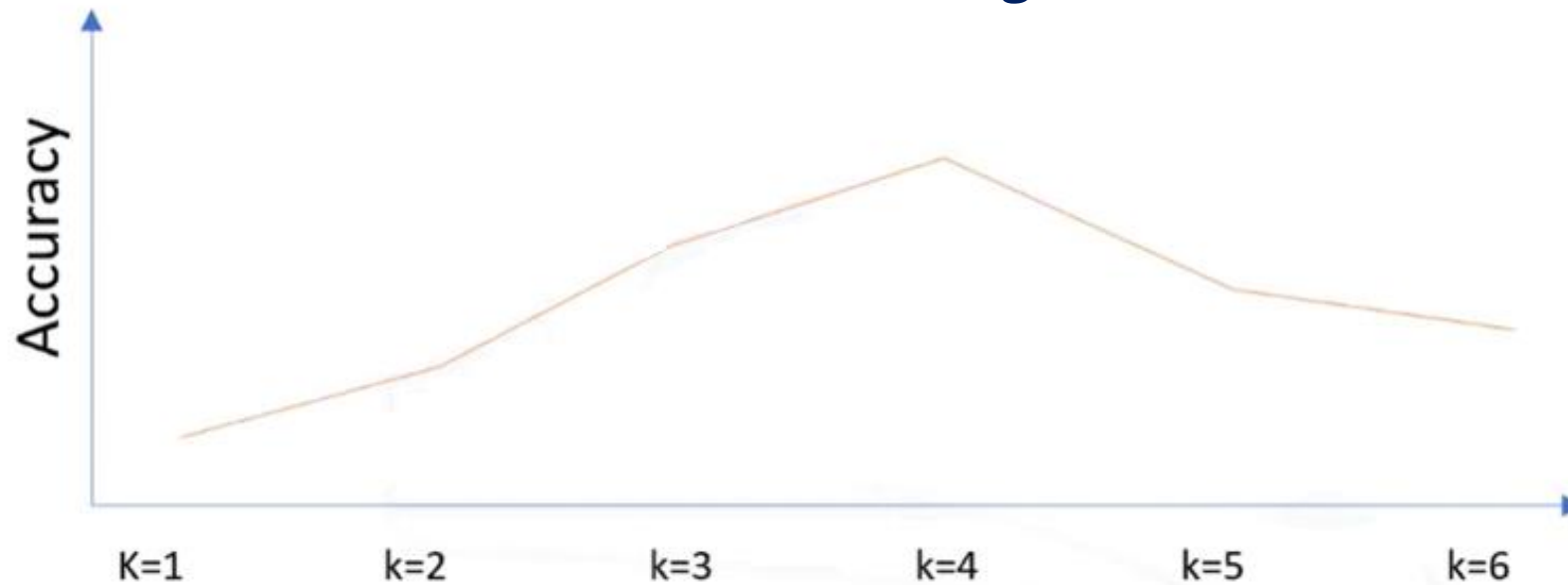
- Supervised Learning
 - KNN



Machine Learning

- Supervised Learning
 - KNN

How to choose the right K?



Machine Learning

- Supervised Learning

Go to menti.com and use the code 1859 7822

What type of supervised algorithm is logistic regression?

0

0

0

Regression

Classification

Clustering

Content

Your question ?

What type of supervised algorithm is logis!

Add longer description

Options ?

Regression

Classification

Clustering

+ Add

Extras

Show correct answer(s) ?

Show results in percentage ?

Let participants submit multiple options

Remove question

LV

Account

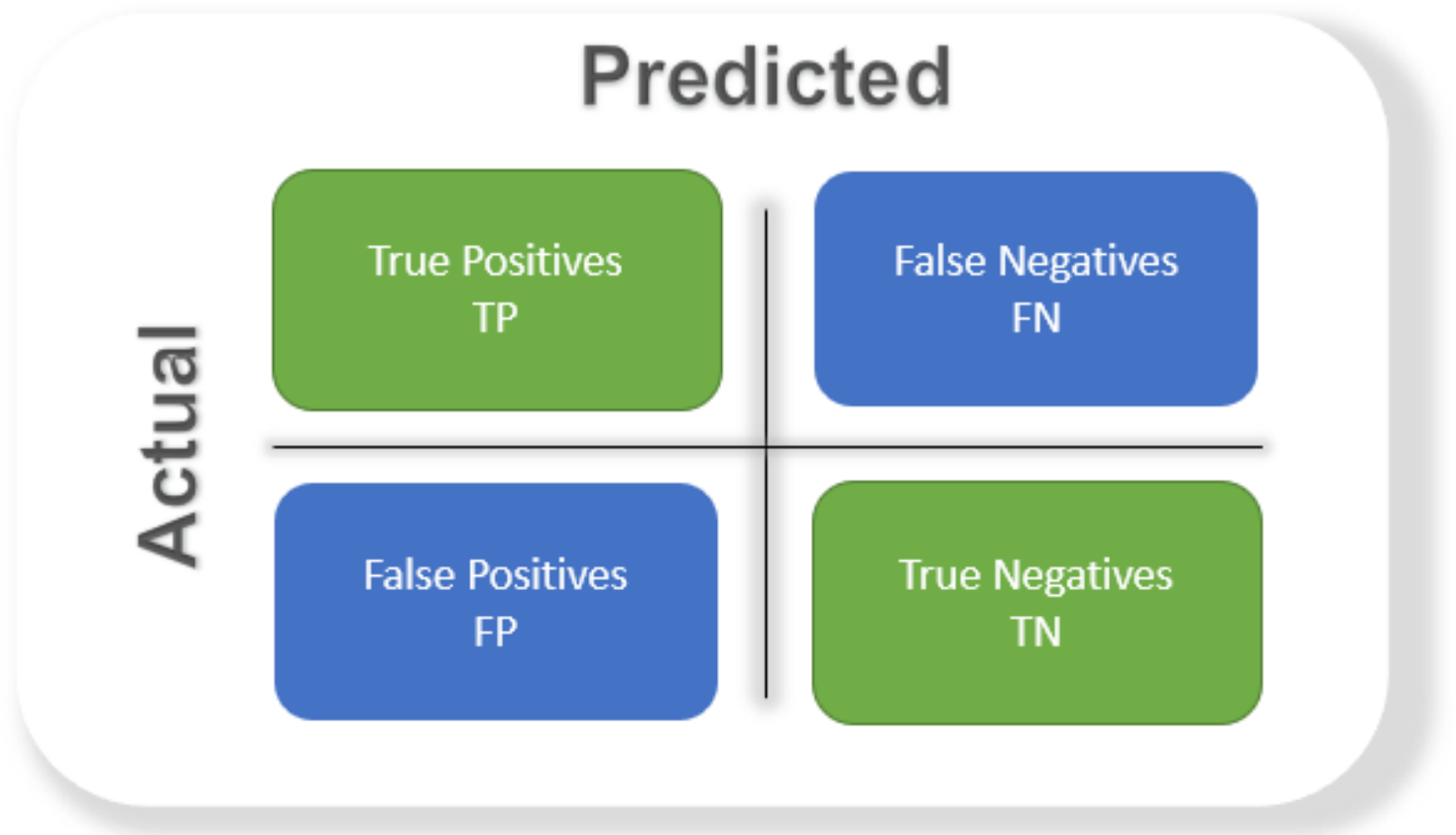
Content

Design

Settings

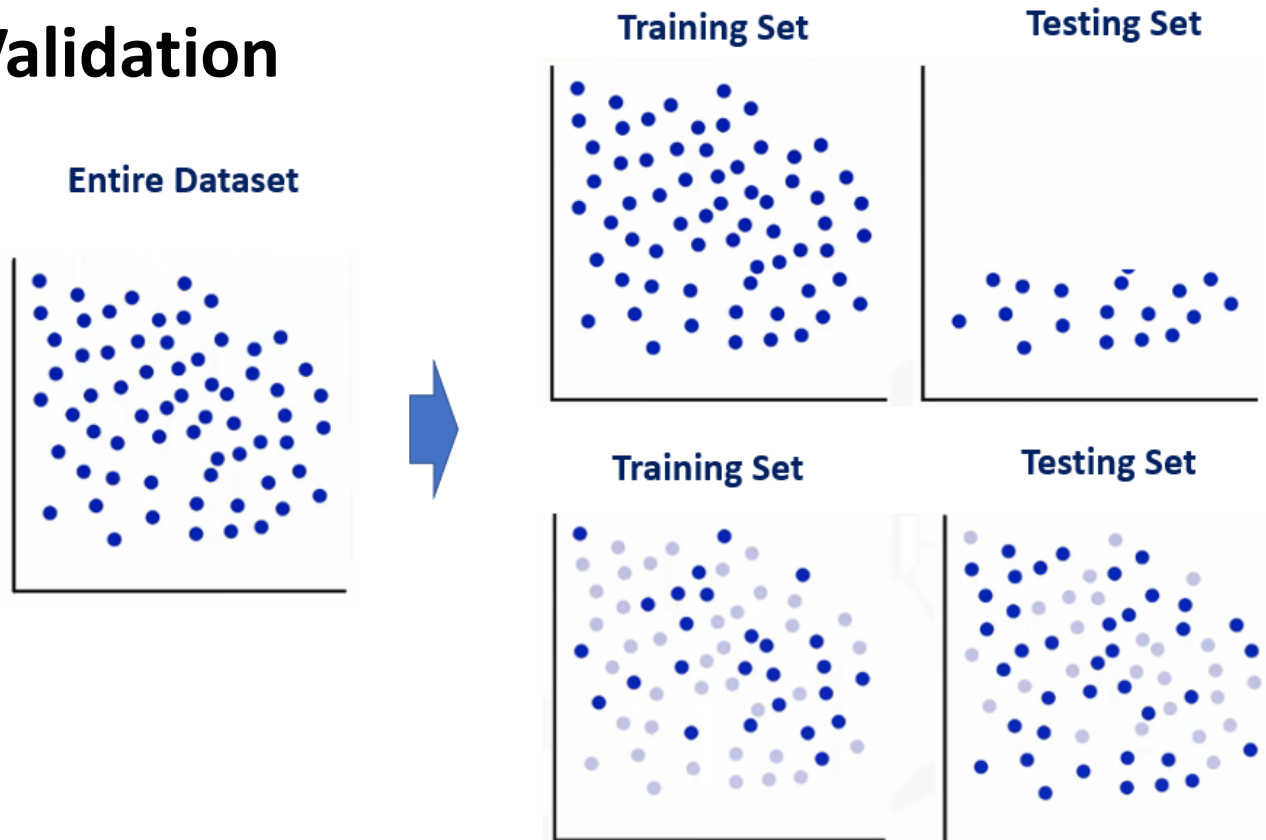
Got feedback?

Model Evaluation



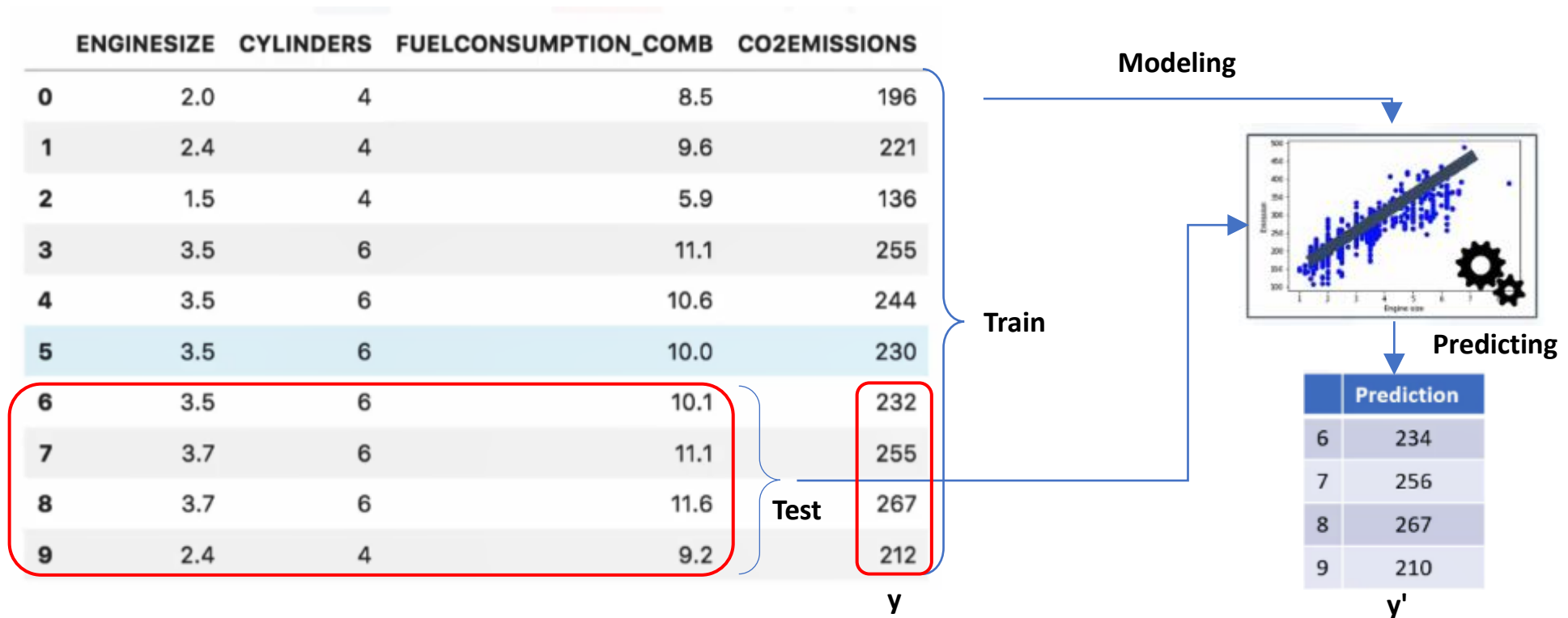
Machine Learning

- **Model Evaluation Approaches**
 - Train and Test on the same Dataset
 - Train/Test Split
 - k-fold Cross Validation



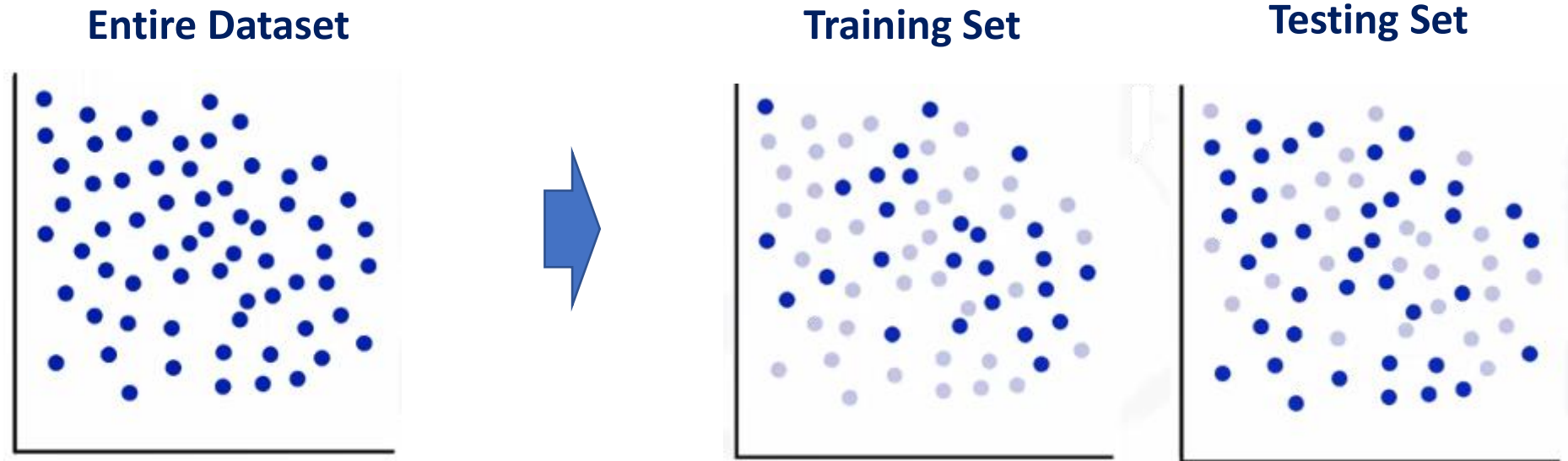
Machine Learning

- Model Evaluation Approaches
 - Train and Test on the same Dataset



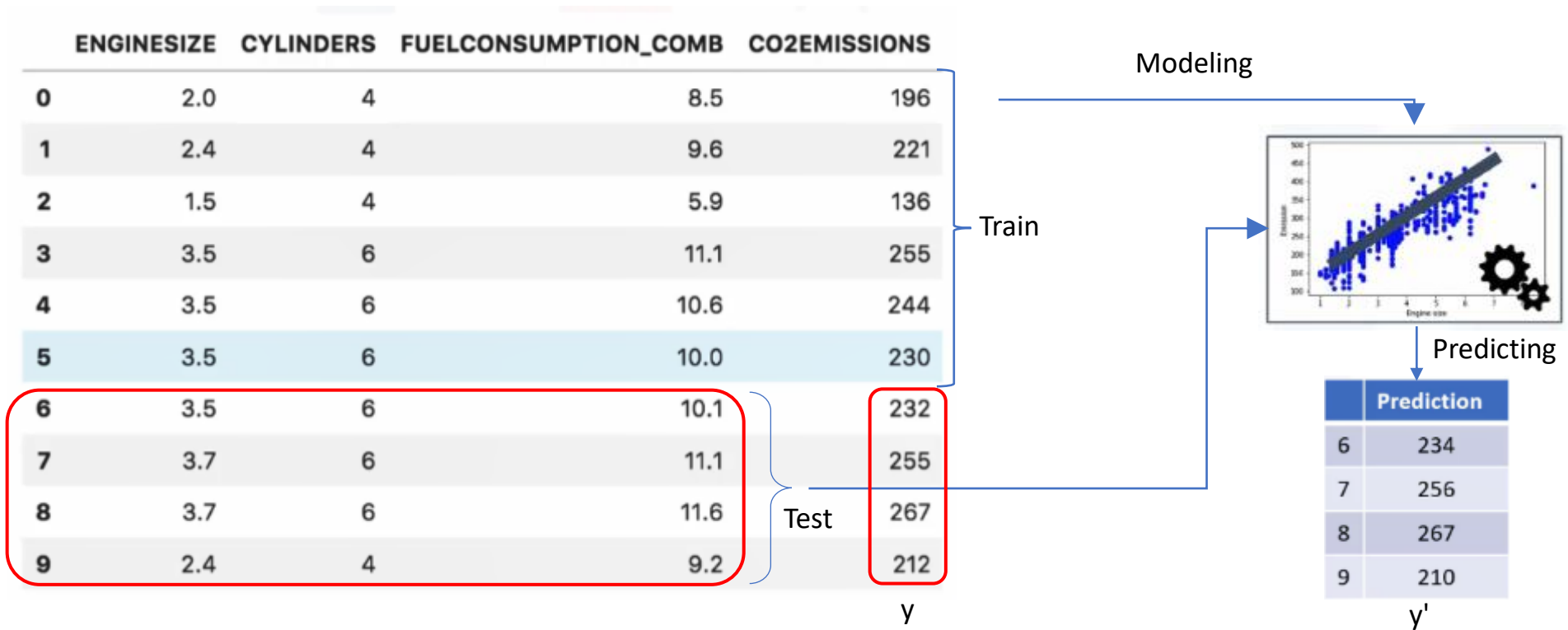
Machine Learning

- Model Evaluation Approaches
 - Train/Test Split



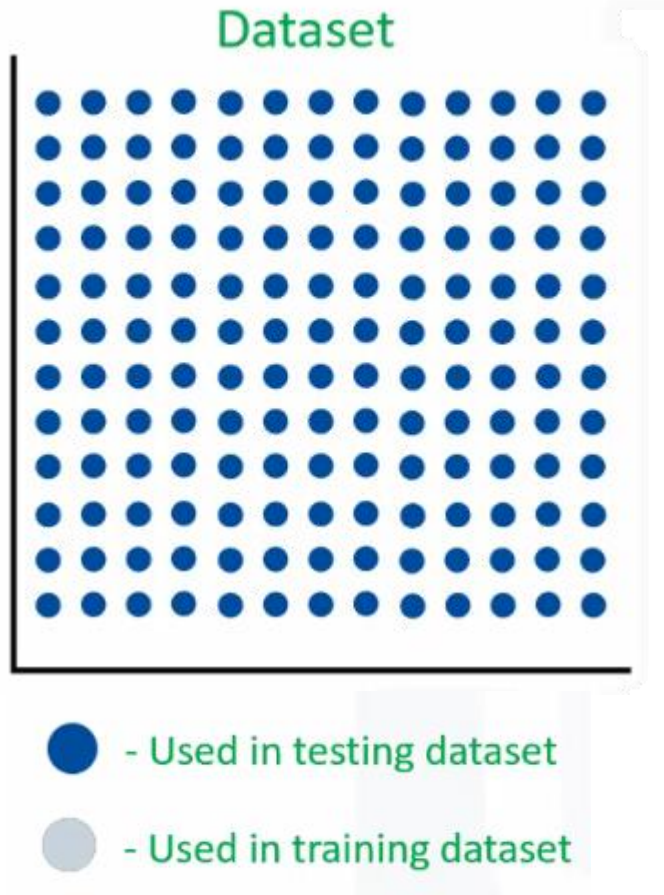
Machine Learning

- Model Evaluation Approaches
 - Train/Test Split

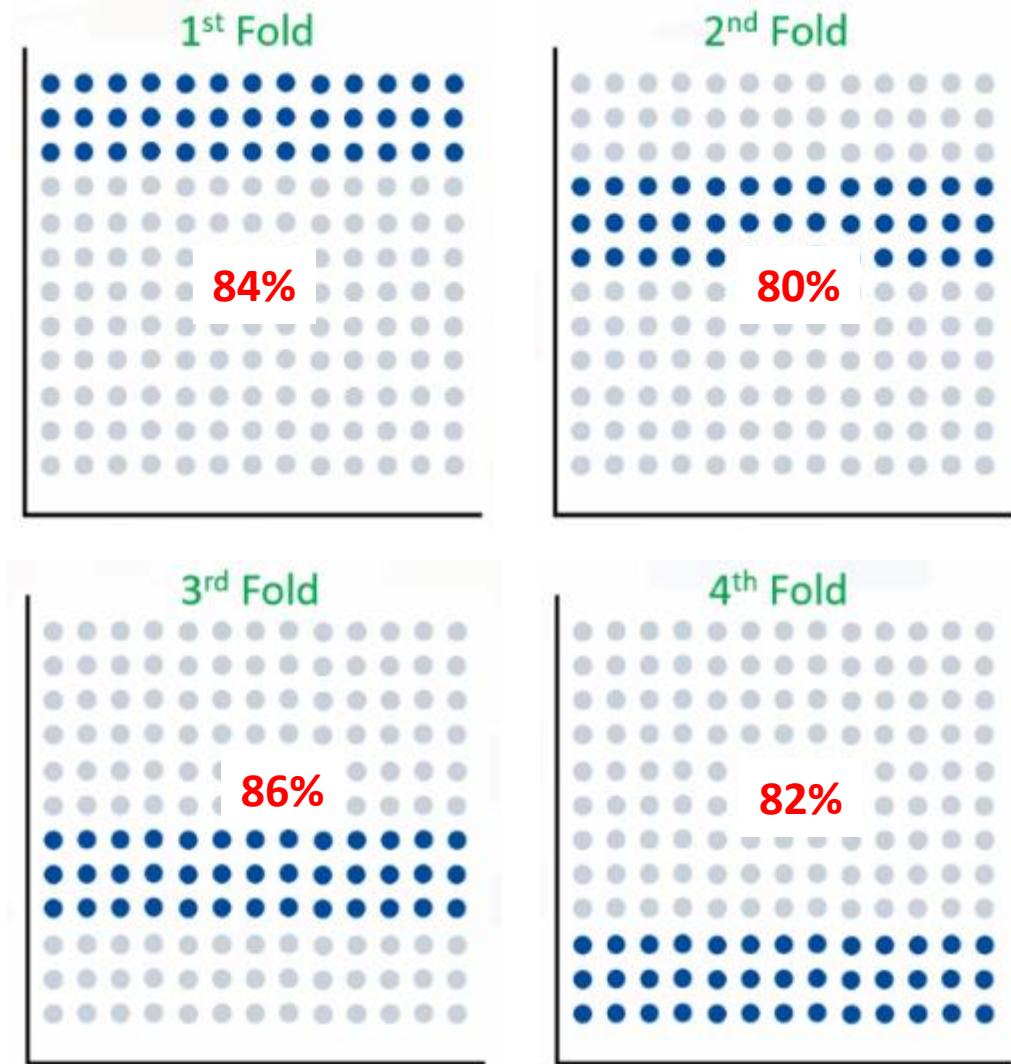


Machine Learning

- Model Evaluation Approaches
 - k-fold Cross Validation

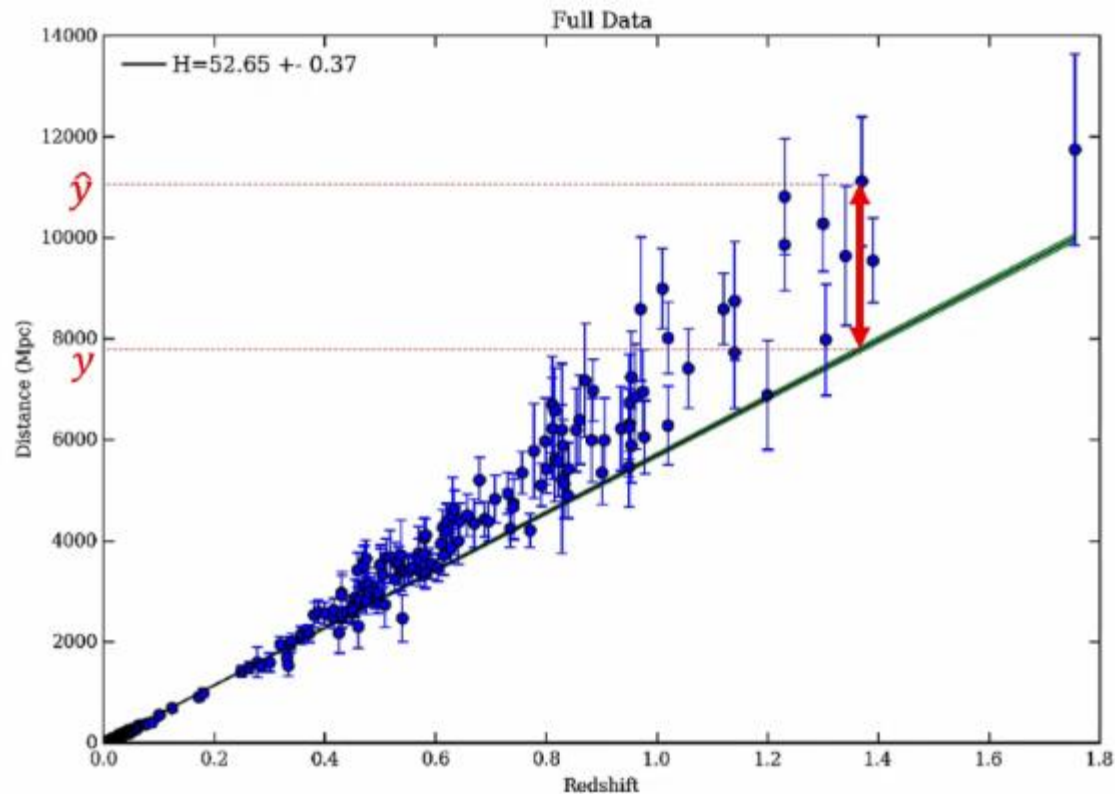


$$\text{Accuracy} = \text{avg} (84\% + 80\% + 86\% + 82\%) = 83\%$$



Machine Learning

- Model Evaluation
 - Evaluation Metrics in Regression



$$MAE = \frac{\sum_i^N |y_i - \hat{y}_i|}{N}$$

Mean Absolute Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Mean Squared Error

$$RMSE = \sqrt{\frac{\sum_1^N (y_i - \hat{y}_i)^2}{N}}$$

Root Mean Squared Error

$$RAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{\sum_{i=1}^n |y_i - \bar{y}|}$$

Relative Absolute Error

$$RSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Relative Squared Error

$$R^2 = 1 - RSE$$

Coefficient of Determination or R-squared

Machine Learning

- **Model Evaluation**

- **Evaluation Metrics in Classification**

- **Accuracy**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**

- **Recall**

- **F1-Score**

$$Precision = \frac{TP}{TP + FP}$$

$$Sensitivity = Recall = \frac{TP}{TP + FN}$$

- **AUC**

$$Recall = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$F1 = 2 * \frac{(precision * recall)}{(precision + recall)}$$

Machine Learning

- Model Evaluation

- Evaluation Metrics in Classification

- Accuracy (Classification Accuracy (CA))

		Actual Values	
		Yes	No
Predicted Values	Yes	True Positive	False Positive
	No	False Negative	True Negative

Confusion Matrix

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- ratio of the number of correct predictions to the total number of input samples
- accuracy is inappropriate for imbalanced classification problems

Machine Learning

- Model Evaluation
 - Evaluation Metrics in Classification
 - Precision

		Actual Values	
		Yes	No
Predicted Values	Yes	True Positive	False Positive
	No	False Negative	True Negative

Confusion Matrix

$$Precision = \frac{TP}{TP + FP}$$

number of positive class predictions that actually belong to the positive class

Machine Learning

- Model Evaluation
 - Evaluation Metrics in Classification
 - Recall

		Actual Values	
		Yes	No
Predicted Values	Yes	True Positive	False Positive
	No	False Negative	True Negative

Confusion Matrix

$$Recall = \frac{TP}{TP + FN}$$

number of positive class predictions made out of all positive examples in the dataset

Machine Learning

- Model Evaluation
 - Evaluation Metrics in Classification
 - F1-score

		Actual Values	
		Yes	No
Predicted Values	Yes	True Positive	False Positive
	No	False Negative	True Negative

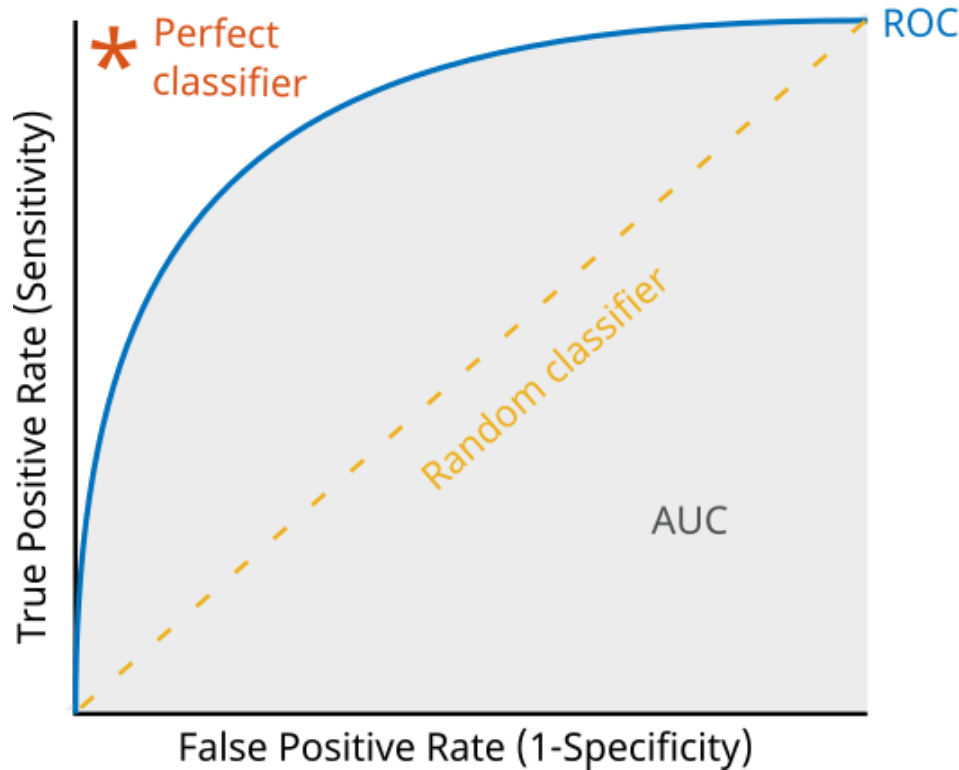
Confusion Matrix

$$F1 = 2 * \frac{(precision * recall)}{(precision + recall)}$$

single score that balances both the concerns of precision and recall in one number

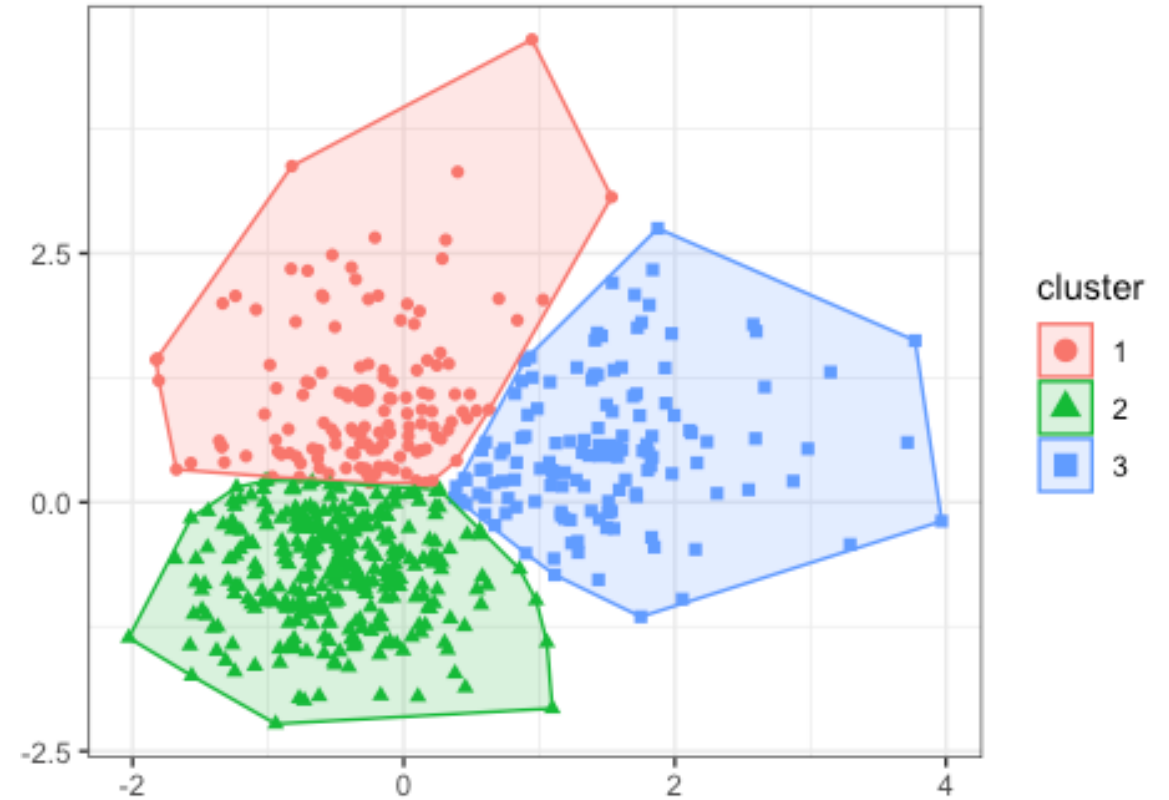
Machine Learning

- Model Evaluation
 - Evaluation Metrics in Classification
 - AUC



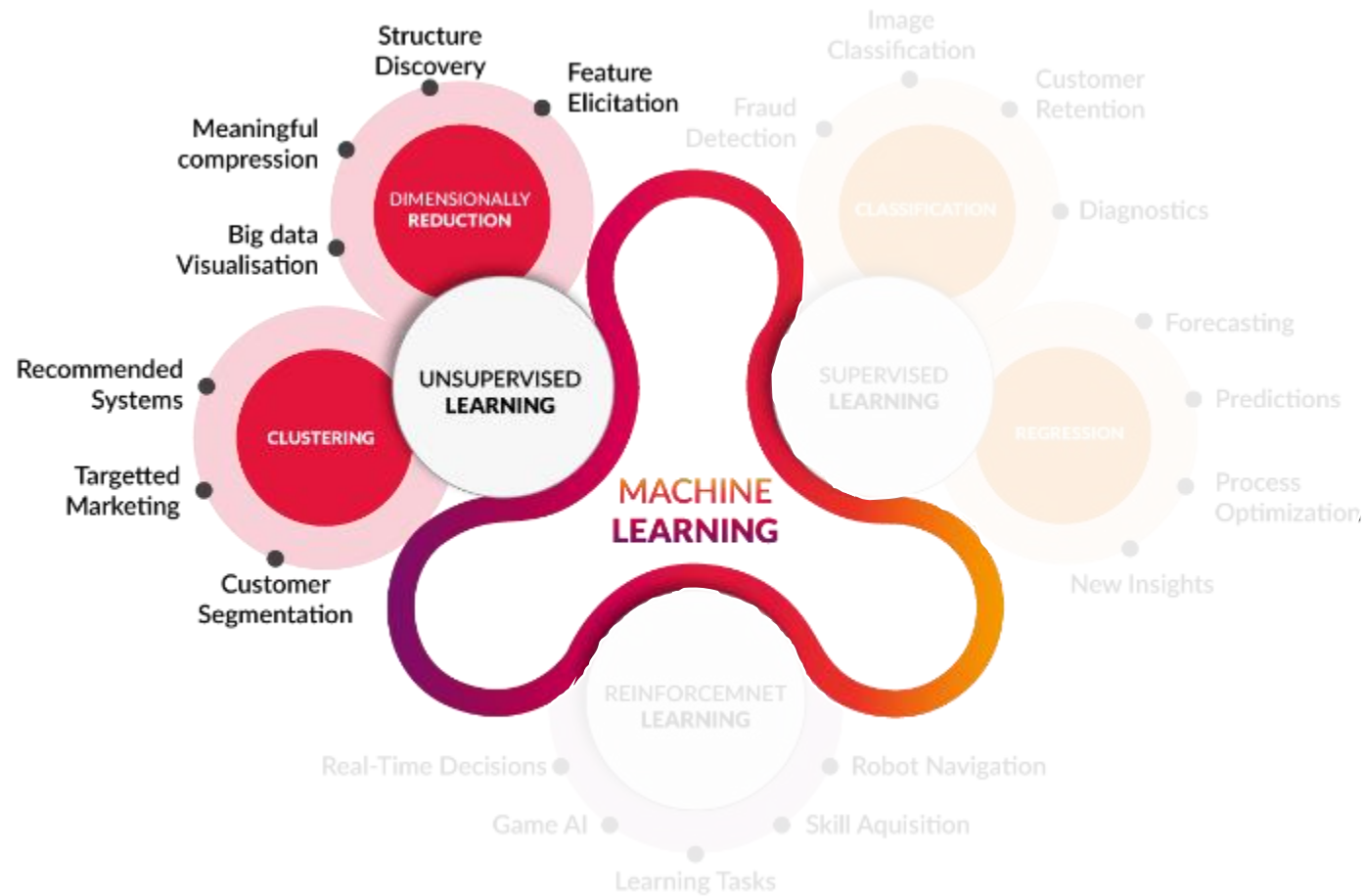
$$\text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN}$$
$$\text{Specificity} = \frac{TN}{TN + FP}$$

Unsupervised Learning



Machine Learning

- Unsupervised Learning



<https://towardsdatascience.com/coding-deep-learning-for-beginners-types-of-machine-learning-b9e651e1ed9d>

Machine Learning

- Unsupervised Learning

Clustering

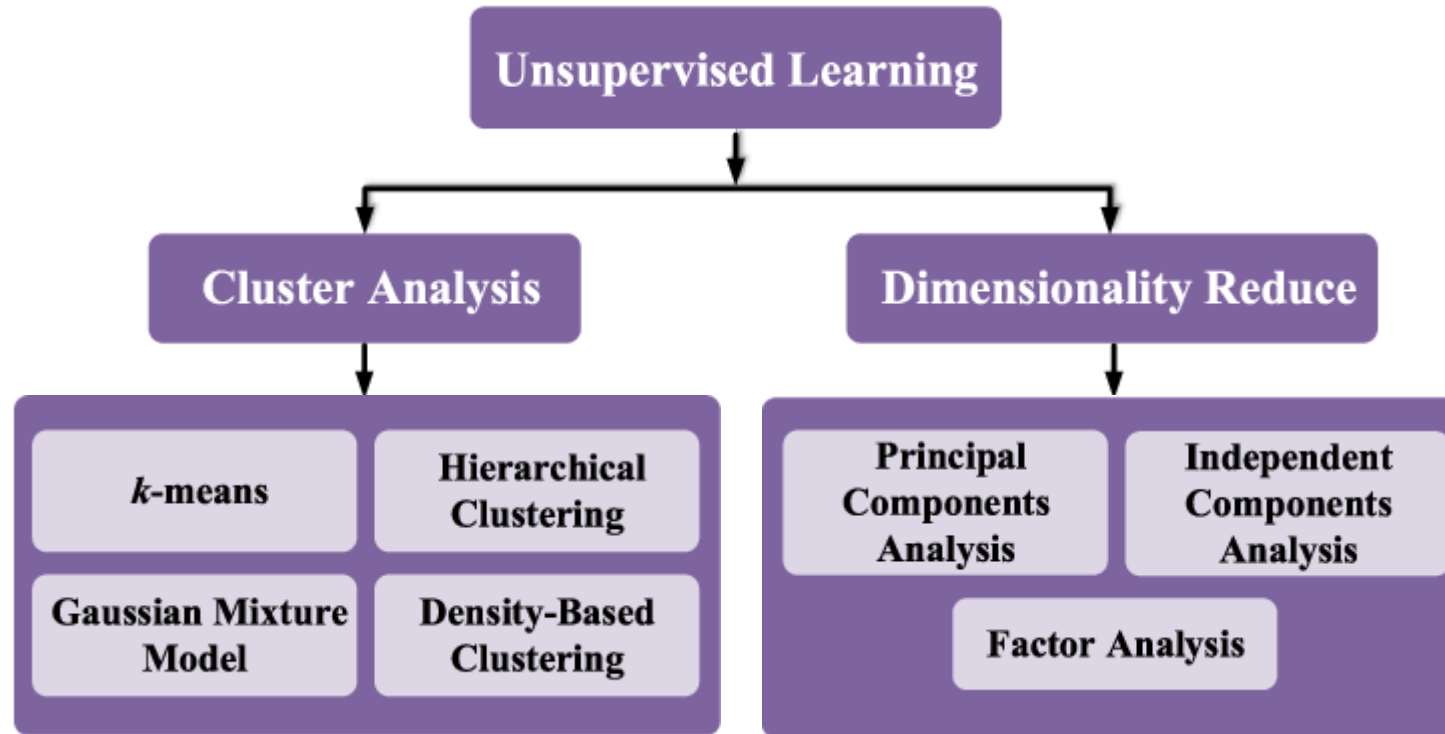
- Process of dividing and **grouping similar data samples together**. Groups are usually called clusters.
- Example usages: segmentation of supermarkets, user base segmentation, signal denoising.

Dimensionality Reduction

- **Data dimension** is the number of features needed to describe data sample.
- **Dimensionality reduction** is a process of compressing features into so-called principal values which conveys similar information concisely.
- Example usages: speeding up other ML algorithms by reducing numbers of calculations, finding a group of most reliable features in data.

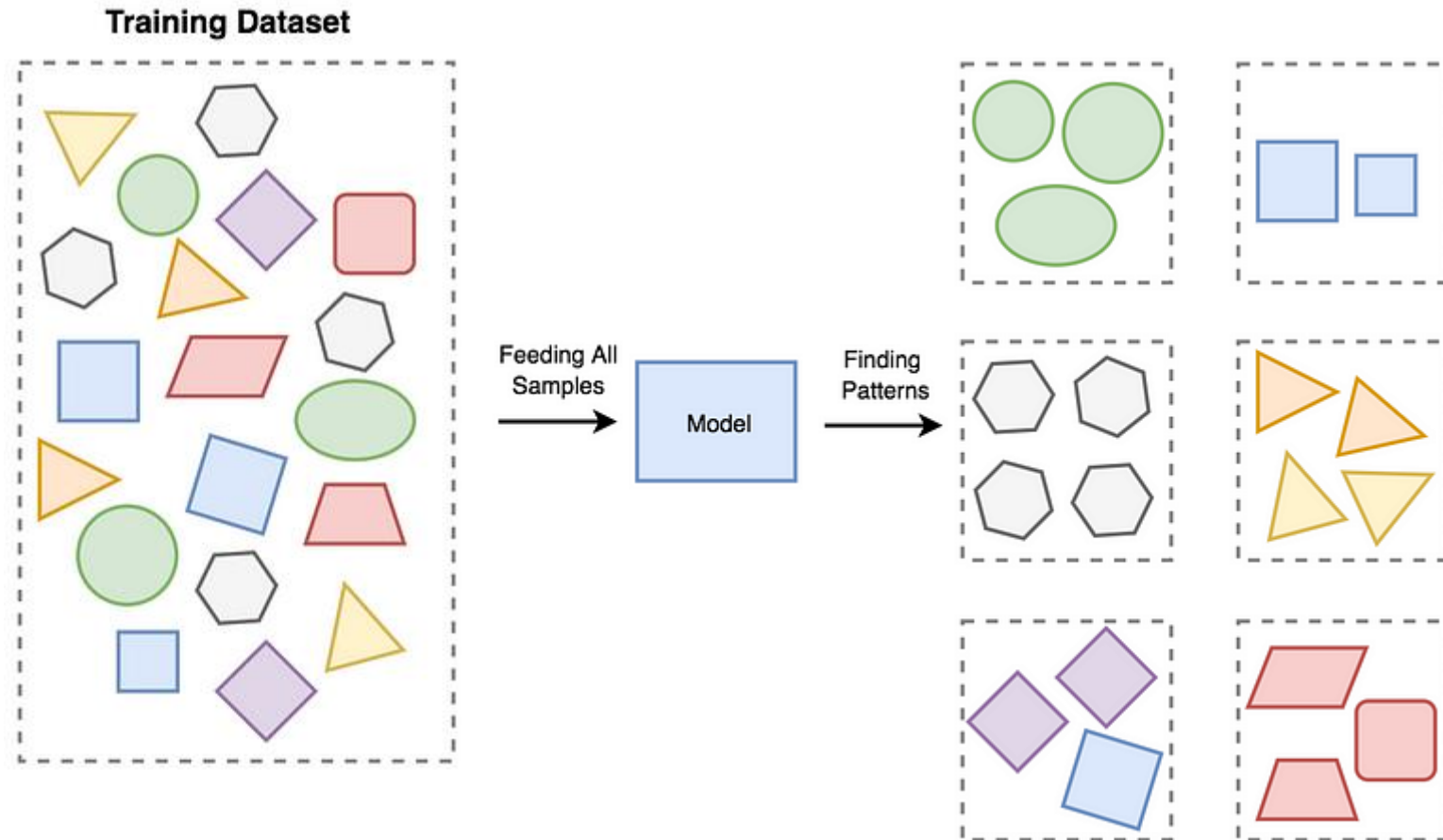
Machine Learning

- Unsupervised Learning



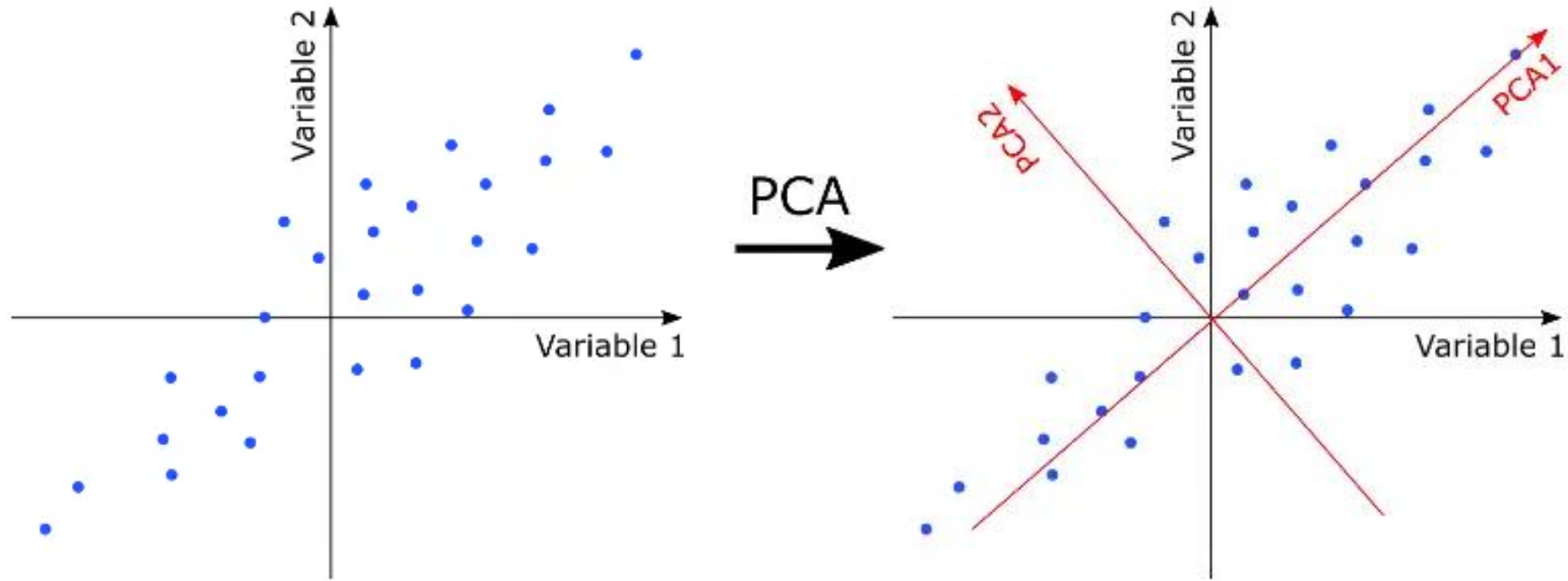
Machine Learning

- Unsupervised Learning
 - Clustering



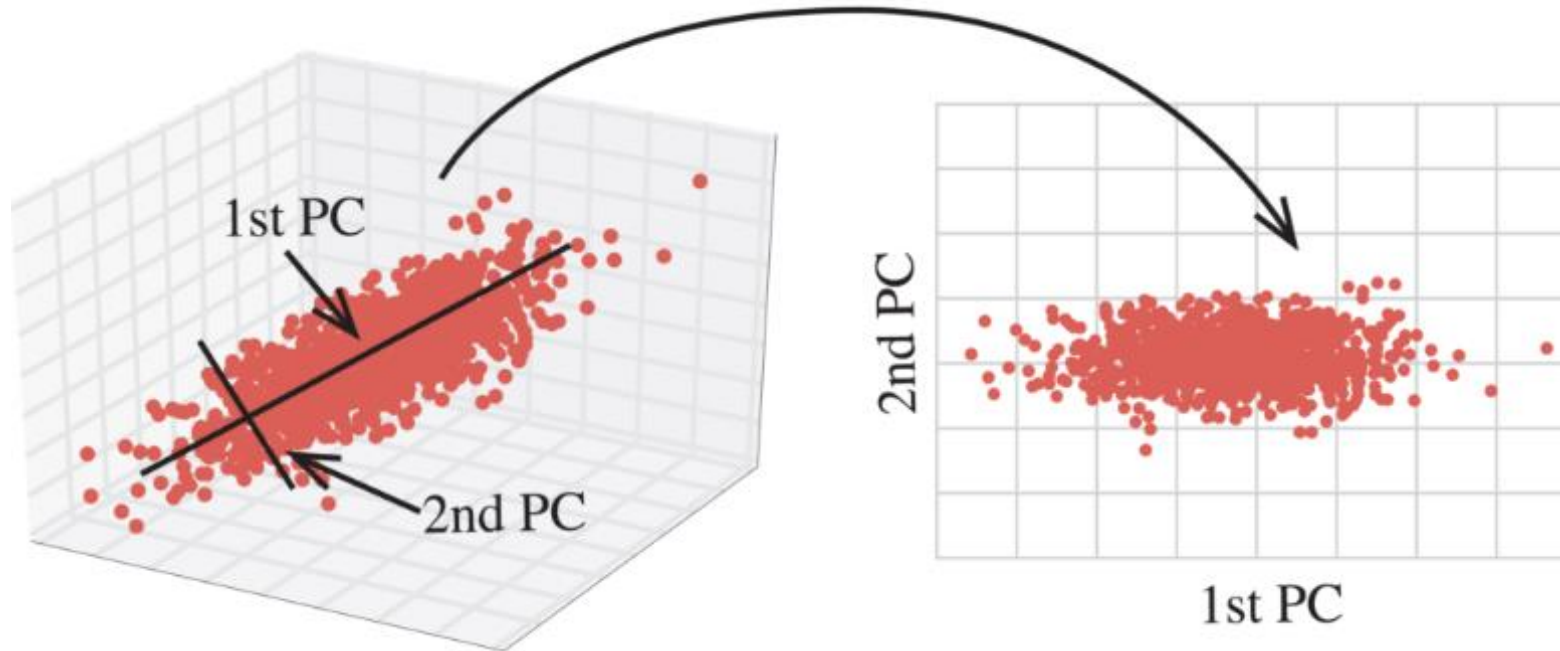
Machine Learning

- Unsupervised Learning
 - Dimensionality Reduction



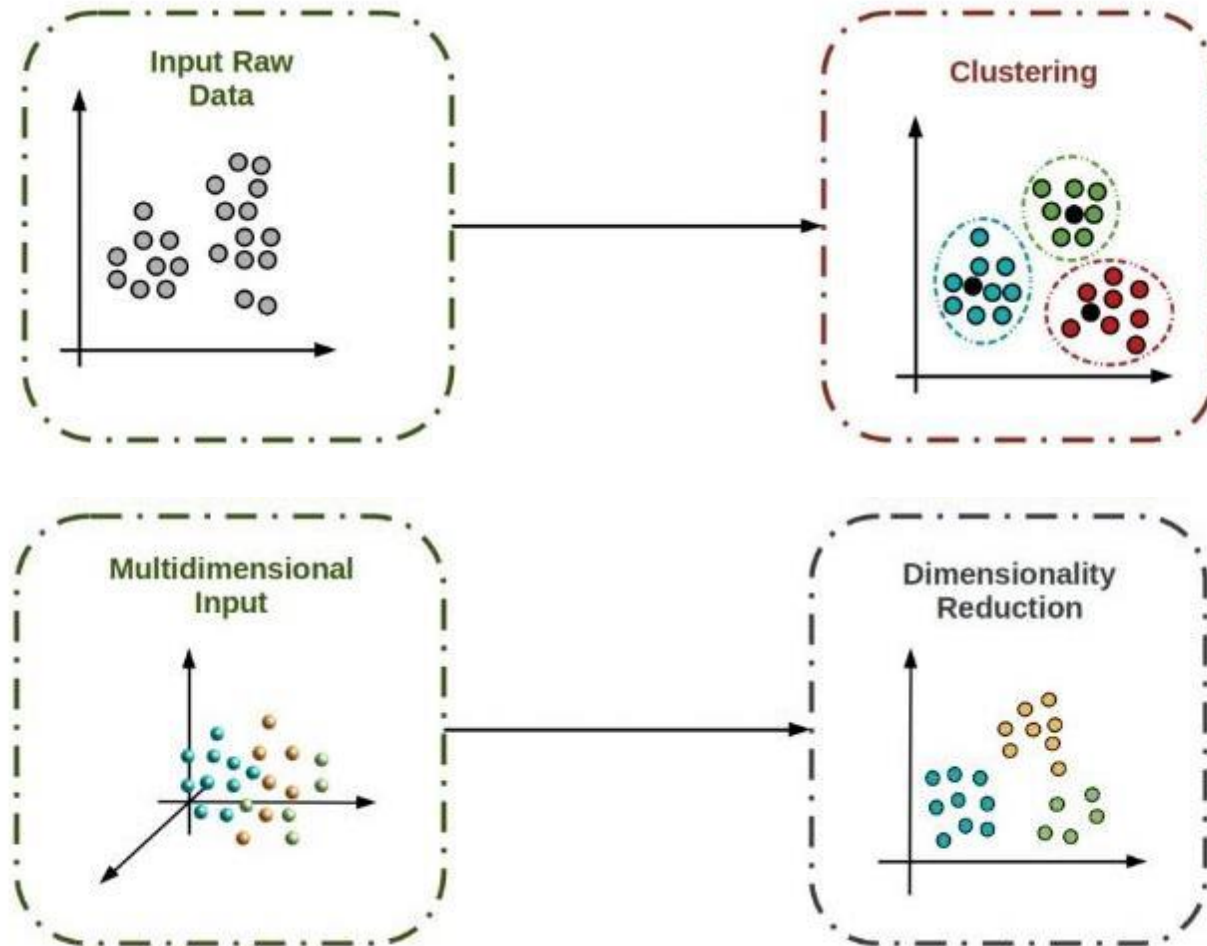
Machine Learning

- Unsupervised Learning
 - Dimensionality Reduction



Machine Learning

- Unsupervised Learning



Clustering

**Dimensionality
Redution**

Tools



Tools

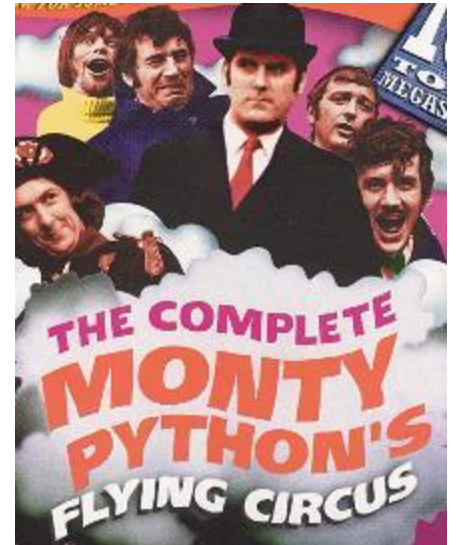
- Overview

	WEKA	KNIME	ORANGE	R	PYTHON	GNU OCTAVE
LOGO						
TOOL TYPE	<u>Data analysis for non-programmers</u>			<u>Data analysis for programmers</u>		
EXTENSIONS / LIBRARIES	Scikit-learn, R, DeepLearning4j, among others	MySQL, H2O, D3.js, Plotly, among others	Python, PostgreSQL, GraphViz, among others	H2O, Plotly, ggplot2, dplyr, among others	Numpy, SciPy, Pandas, Matplotlib, among others	Scientific Library, Dionysus, nurbs, geometry, among others
PRICE	Free	Free / Different pay versions	Free			

<https://datos.gob.es/en/blog/most-popular-data-analysis-tools>

Tools

- **Python**
- high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation via the off-side rule.
- the name of the Python programming language comes from an old BBC television comedy sketch series called Monty Python's Flying Circus



Tools

- **Python**
- **Libraries**
 - **Scikit-Learn:** implements many Machine Learning algorithms. Created by David Cournapeau in 2007 is now lead by researchers from Inria
 - **TensorFlow:** machine learning library that permits distributed numerical computation. It was created at Google. It was open sourced in November 2015.
 - **Keras:** high level Deep Learning library that makes that can run on top of either TensorFlow, Theano or Microsoft Cognitive Toolkit (formerly known as CNTK).



Keras

Tools

- Python
- IDEs



PyCharm



Visual Studio Code



Sublime Text



Vim



GNU Emacs



Spyder



Atom



Jupyter



IntelliJ IDEA

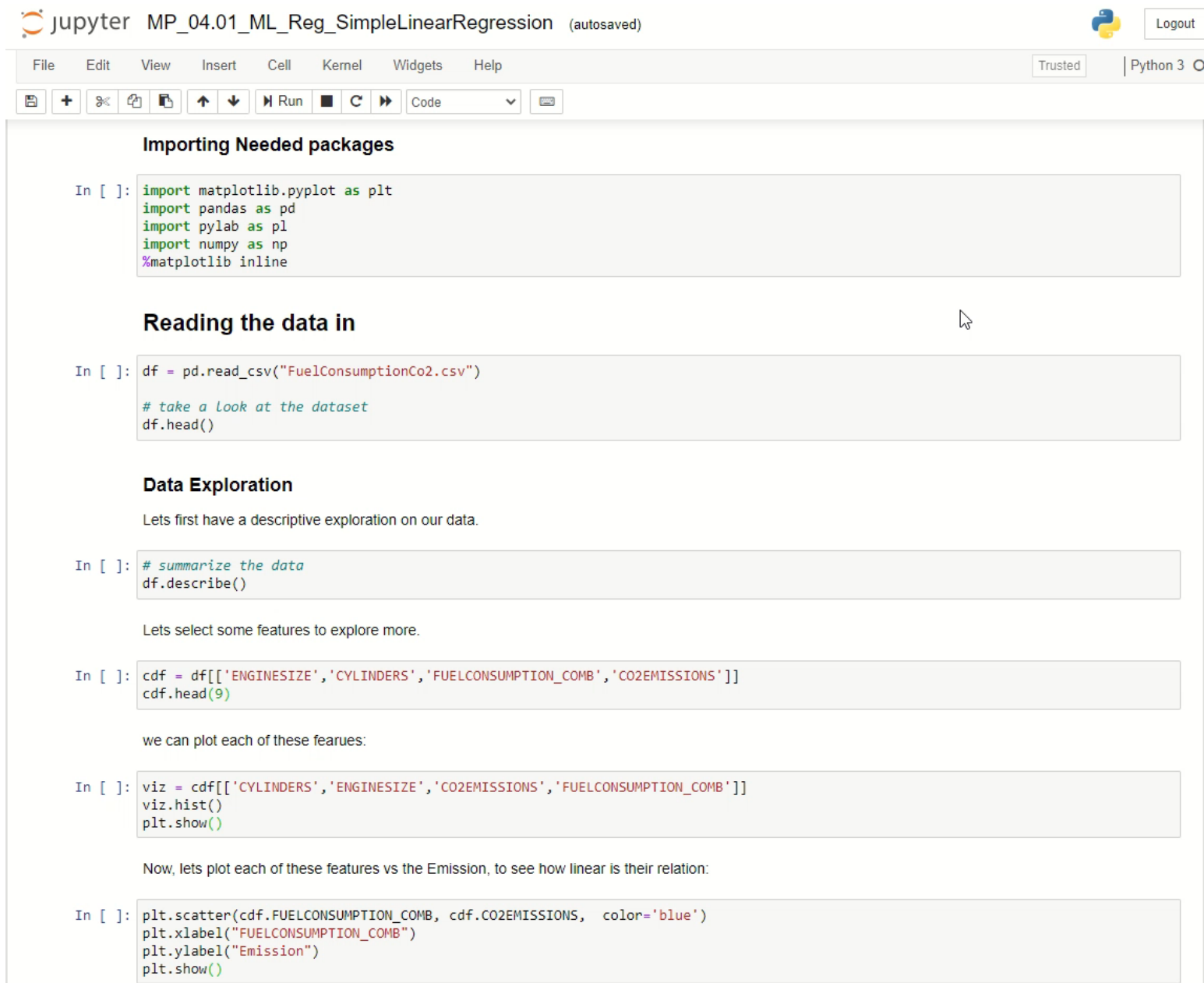


Notepad++



Tools

- Python
- IDEs



The image shows a Jupyter Notebook interface with the title "MP_04.01_ML_Reg_SimpleLinearRegression (autosaved)". The interface includes a top menu bar with options like File, Edit, View, Insert, Cell, Kernel, Widgets, and Help. Below the menu is a toolbar with icons for saving, adding cells, undo, redo, and running code. The notebook content is organized into sections:

- Importing Needed packages**: A code cell containing imports for matplotlib.pyplot (plt), pandas (pd), pylab (pl), numpy (np), and an in-line matplotlib directive.
- Reading the data in**: A code cell that reads a CSV file named "FuelConsumptionCo2.csv" into a DataFrame (df) and displays the first few rows.
- Data Exploration**: A section with a text prompt "Lets first have a descriptive exploration on our data." followed by a code cell using df.describe() to summarize the data.
- A text prompt "Lets select some features to explore more." is followed by a code cell selecting specific columns (ENGINE SIZE, CYLINDERS, FUEL CONSUMPTION_COMB, CO2 EMISSIONS) and displaying the first 9 rows.
- A text prompt "we can plot each of these fearues:" (note the typo) is followed by a code cell creating a visualization object (viz) and displaying histograms for the selected features.
- A text prompt "Now, lets plot each of these features vs the Emission, to see how linear is their relation:" is followed by a code cell creating a scatter plot of FUEL CONSUMPTION_COMB vs Emission, colored blue.

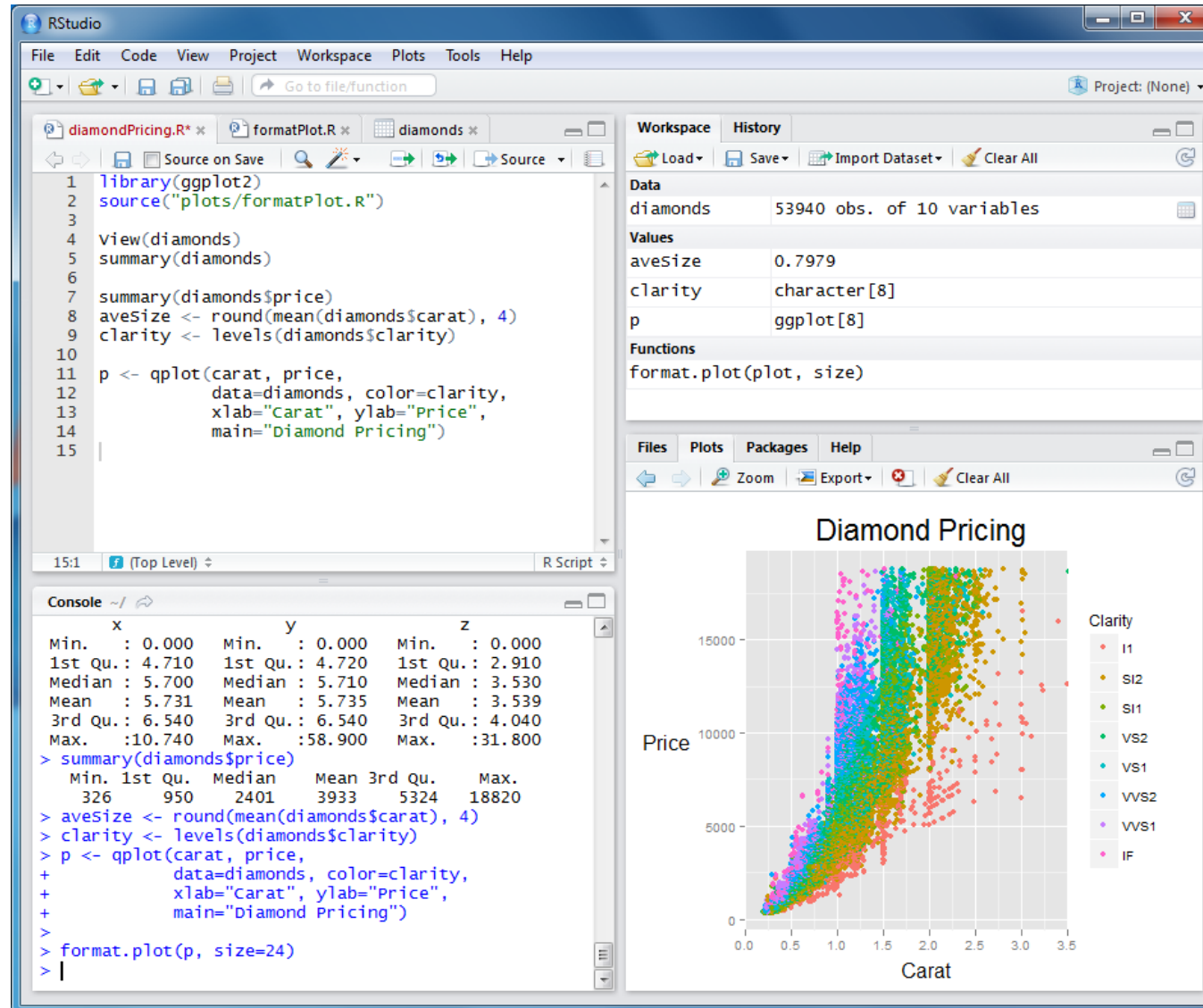
Tools

- R
 - a programming language for statistical computing and graphics
 - open-source free software environment within the GNU package and under the GNU General Public License.
 - R has a command line interface.
 - Multiple third-party graphical user interfaces are also available
 - RStudio, an integrated development environment (IDE)
 - Jupyter, a notebook interface.



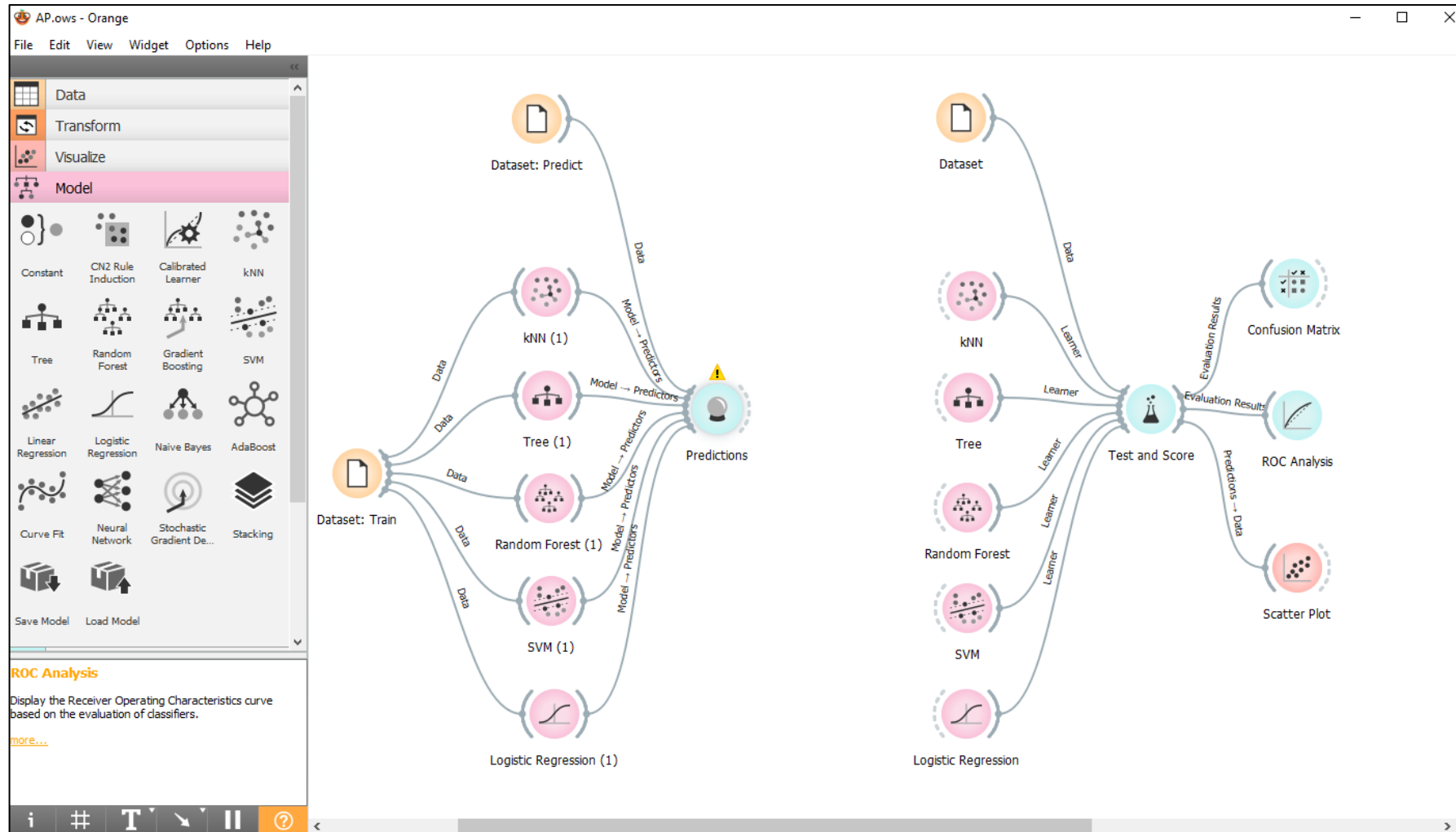
Tools

- R



Tools

- Orange



Tools

- Orange



This is the description of the features of the dataset.

1. Sample code number: id number
2. Clump Thickness: 1 - 10
3. Uniformity of Cell Size: 1 - 10
4. Uniformity of Cell Shape: 1 - 10
5. Marginal Adhesion: 1 - 10
6. Single Epithelial Cell Size: 1 - 10
7. Bare Nuclei: 1 - 10
8. Bland Chromatin: 1 - 10
9. Normal Nucleoli: 1 - 10
10. Mitoses: 1 - 10
11. Class: (2 for benign, 4 for malignant)

<https://www.kaggle.com/code/karteek93/breast-cancer-classification/notebook>

Tools

- Orange

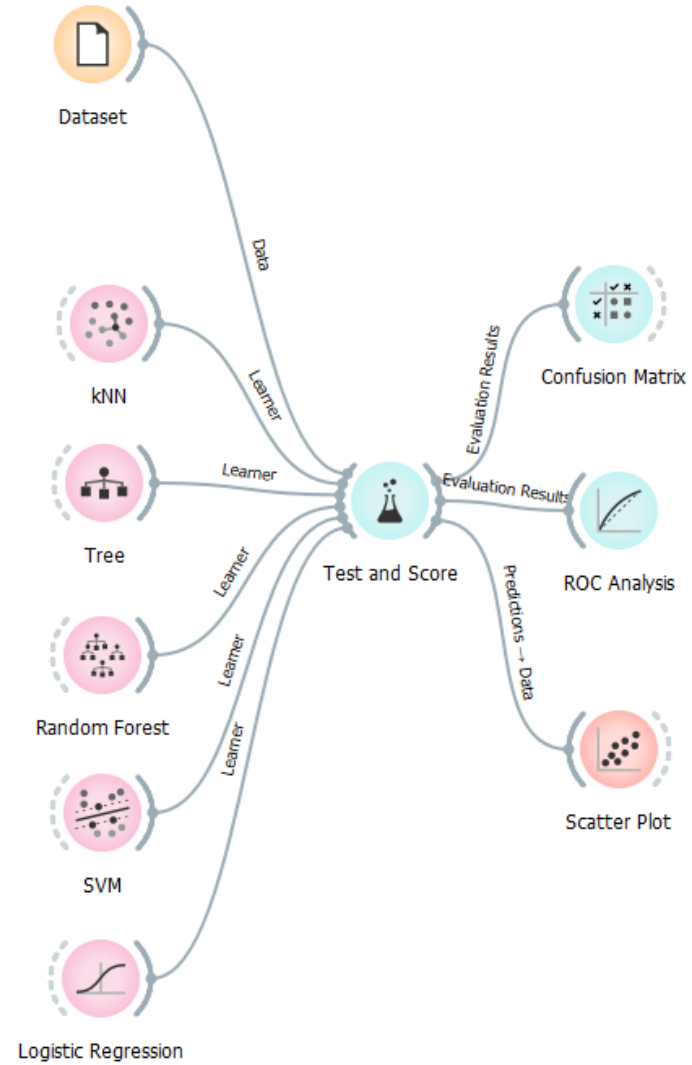
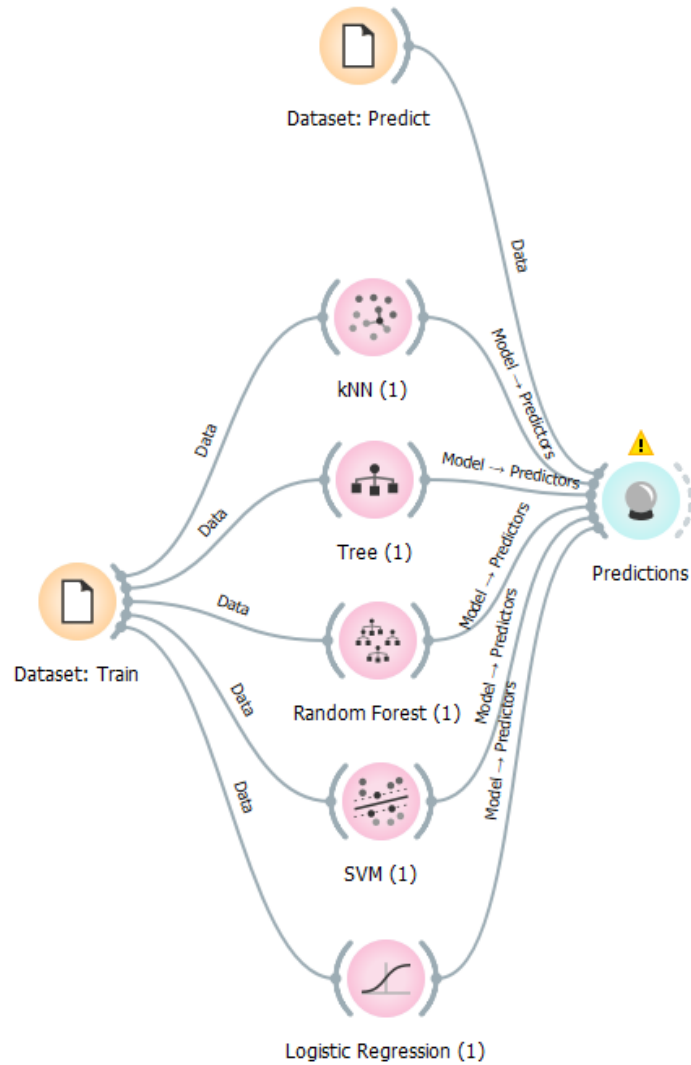


	A	B	C	D	E	F	G	H	I	J	K
1	ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSiz	BareNuc	BlandChrc	NormNucl	Mit	Class
2	1000025	5	1	1	1	2	1	3	1	1	2
3	1002945	5	4	4	5	7	10	3	2	1	2
4	1015425	3	1	1	1	2	2	3	1	1	2
5	1016277	6	8	8	1	3	4	3	7	1	2
6	1017023	4	1	1	3	2	1	3	1	1	2
7	1017122	8	10	10	8	7	10	9	7	1	4
8	1018099	1	1	1	1	2	10	3	1	1	2
9	1018561	2	1	2	1	2	1	3	1	1	2
10	1033078	2	1	1	1	2	1	1	1	5	2
11	1033078	4	2	1	1	2	1	2	1	1	2
12	1035283	1	1	1	1	1	1	3	1	1	2
13	1036172	2	1	1	1	2	1	2	1	1	2
14	1041801	5	3	3	3	2	3	4	4	1	4
15	1043999	1	1	1	1	2	3	3	1	1	2
16	1044572	8	7	5	10	7	9	5	5	4	4
17	1047630	7	4	6	4	6	1	4	3	1	4
18	1048672	4	1	1	1	2	1	2	1	1	2
19	1049815	4	1	1	1	2	1	3	1	1	2
20	1050670	10	7	7	6	4	10	4	1	2	4
21	1050718	6	1	1	1	2	1	3	1	1	2

<https://www.kaggle.com/code/karteeek93/breast-cancer-classification/notebook>

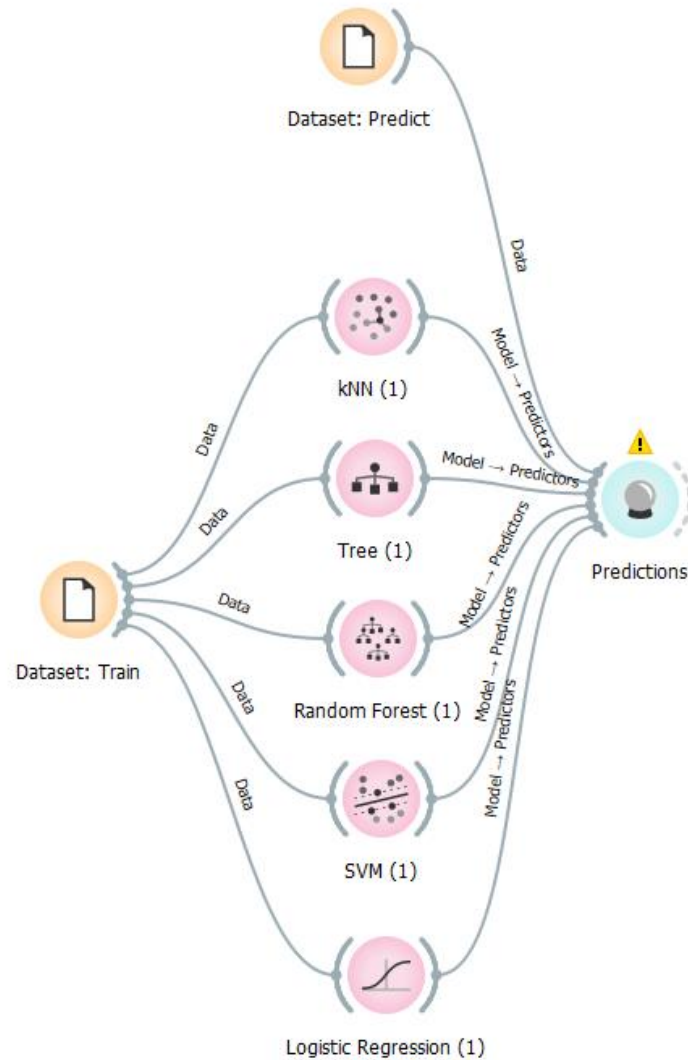
Tools

- Orange



Tools

- Orange



Predictions - Orange

Show probabilities for: **Classes in data**

Restore Original Order

	kNN	Tree (1)	Random Forest (1)	SVM (1)	Logistic Regression (1)	Class	ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize
1	2	2	2	2	2	?	5000001	5	1	1	1	2
2	2	2	4	2	2	?	5000002	5	4	4	5	7
3	2	2	2	2	2	?	5000003	3	1	1	1	2
4	2	4	2	2	2	?	5000004	6	8	8	1	3
5	2	2	2	2	2	?	5000005	4	1	1	3	2
6	2	2	2	2	2	?	5000006	1	1	1	1	2
7	2	2	2	2	2	?	5000007	2	1	2	1	2
8	2	2	2	2	2	?	5000008	2	1	1	1	2
9	2	2	2	2	2	?	5000009	4	2	1	1	2
10	2	2	2	2	2	?	5000010	1	1	1	1	1

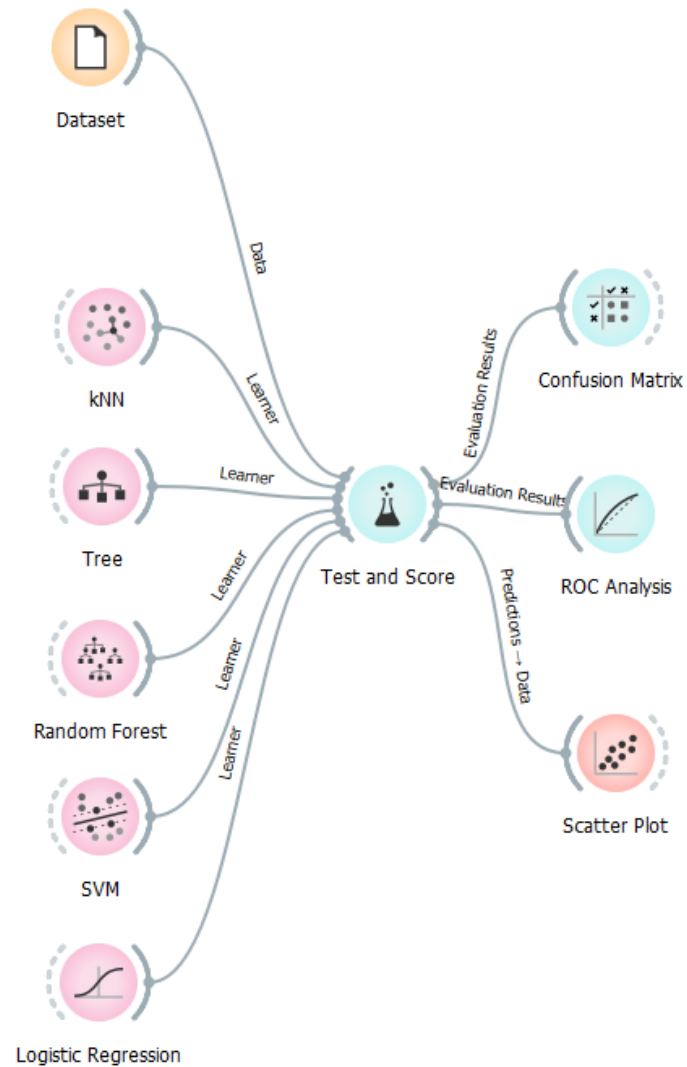
✓ Show performance scores Target class: (Average over classes)

Model	AUC	CA	F1	Precision	Recall
kNN	NA	NA	NA	NA	NA
Tree (1)	NA	NA	NA	NA	NA
Random Forest (1)	NA	NA	NA	NA	NA
SVM (1)	NA	NA	NA	NA	NA
Logistic Regression (1)	NA	NA	NA	NA	NA

Instances with missing targets are ignored while scoring.

Tools

- Orange



Test and Score - Orange

☒ Cross validation
Number of folds: 10
☒ Stratified

☐ Cross validation by feature

☐ Random sampling
Repeat train/test: 20
Training set size: 70 %
☒ Stratified

☐ Leave one out
☐ Test on train data
☐ Test on test data

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Precision	Recall
kNN	0.551	0.597	0.580	0.571	0.597
Tree	0.933	0.938	0.938	0.938	0.938
SVM	0.965	0.883	0.877	0.893	0.883
Random Forest	0.991	0.964	0.964	0.965	0.964
Logistic Regression	0.558	0.655	0.519	0.429	0.655

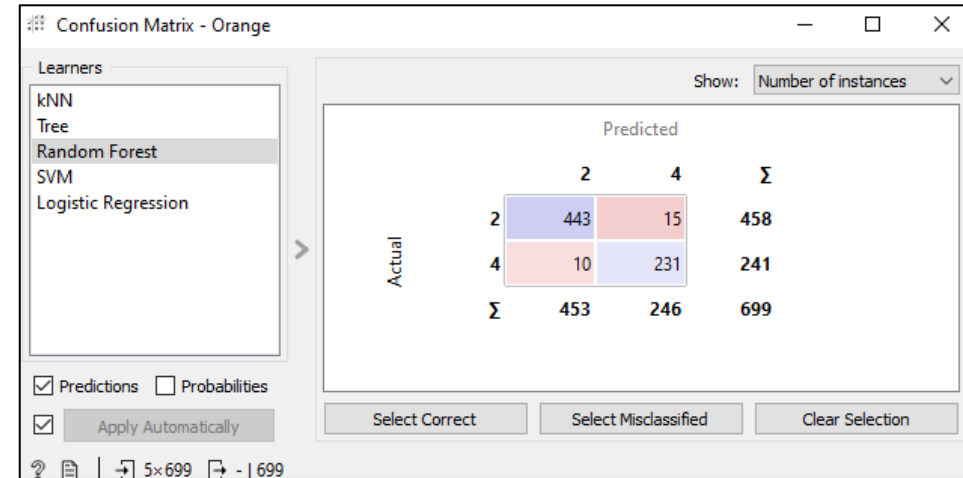
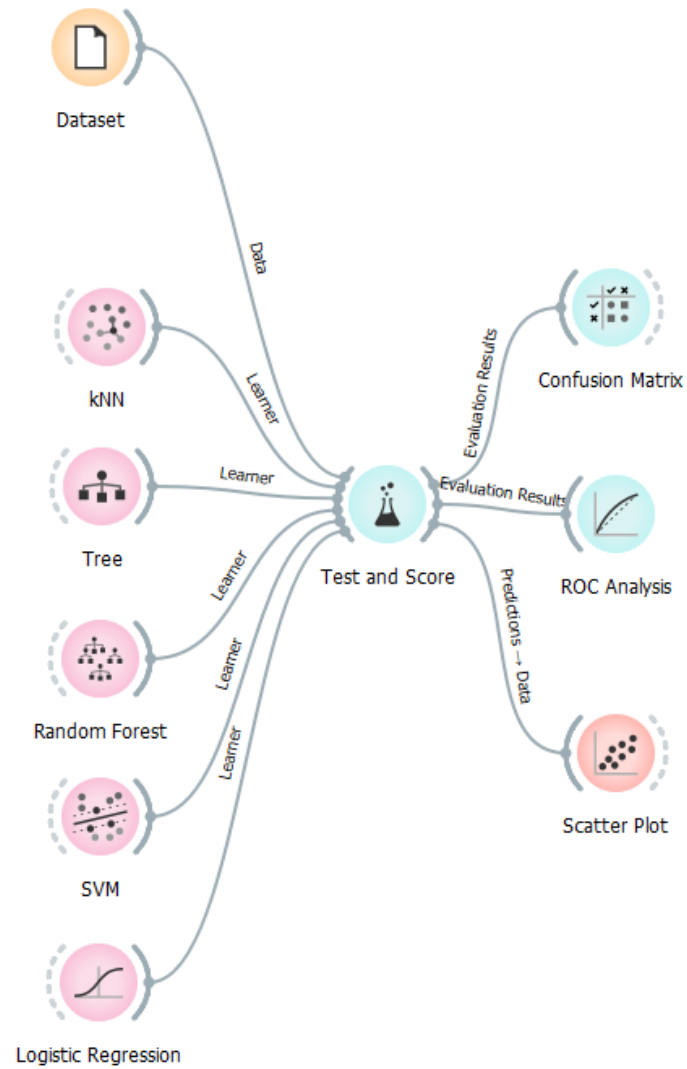
Compare models by: Area under ROC curve
☒ Negligible diff.: 0.01

	kNN	Tree	SVM	Random Forest	Logistic Regression
kNN		0.000 0.000	0.000 0.000	0.000 0.000	0.365 0.156
Tree	1.000 0.000		0.018 0.094	0.001 0.004	1.000 0.000
SVM	1.000 0.000	0.888 0.094		0.007 0.105	1.000 0.000
Random Forest	1.000 0.000	0.996 0.004	0.888 0.105		1.000 0.000
Logistic Regression	0.479 0.156	0.000 0.000	0.000 0.000	0.000 0.000	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

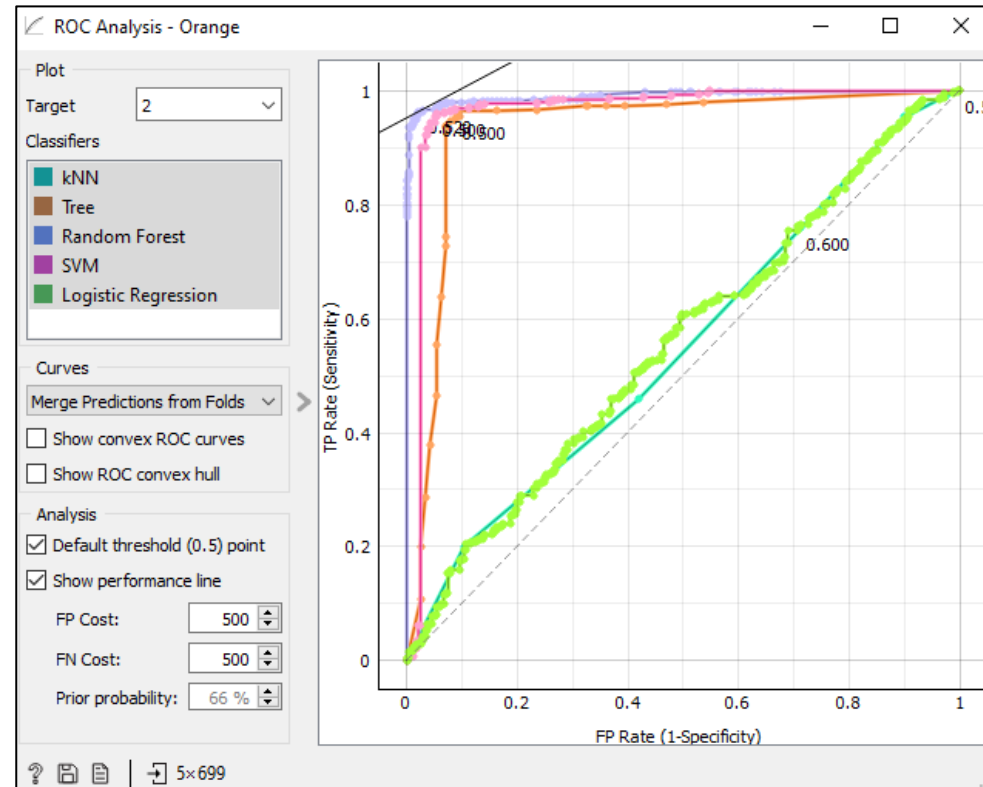
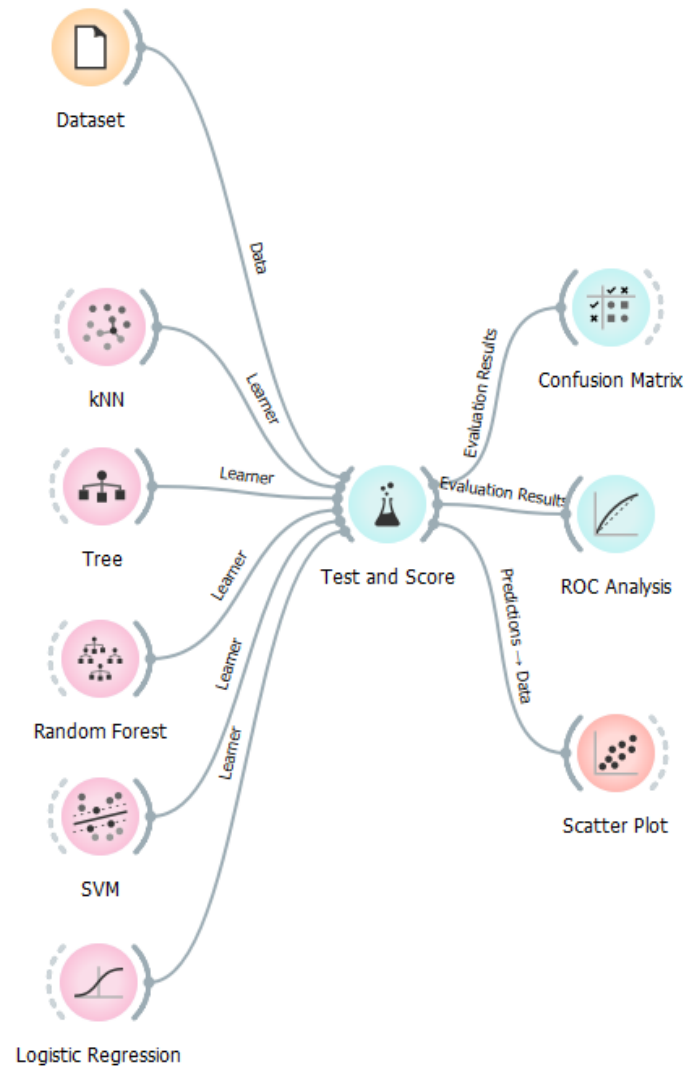
Tools

- Orange



Tools

- Orange



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